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Machine learning-based digital district heating/cooling with renewable integrations and advanced low-carbon transition

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A R T I C L E I N F O

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Resilient and smart grids' interactions

A B S T R A C T

Intermittent power production with hybrid storages, dynamic grids' interactions for synergistic complementation, advanced energy management, optimal design and robust operation are critical approaches to realise smart district energy systems. Inter-city energy migration framework with energy flexibility can improve efficiency and enhance resilience in response to fluctuations in power supply and demand. However, limited studies focused on up-to-date technology advances and artificial intelligence-assisted control for district energy systems. This study comprehensively reviewed district heating and cooling networks with diversified grids' interactions, smart energy management and control strategy through multi-disciplinary approaches. An inter-city transportation-based energy migration framework was proposed for district energy sharing and regional energy balance. Technical feasibility and prospects of machine learning methods on energy planning and optimisation have also been demonstrated in terms of demand prediction, energy dispatch, surrogate model development for uncertainty analysis and optimisation, geometrical and operating parameter design. A district energy network was formulated, involving on-site renewable generations, waste heat recovery from centralized power plants, multi-diversified energy storages, advanced energy conversions for distributed renewable energy sharing. Several technical challenges were identified as avenues for future research, including benchmarks for selection of most suitable energy storages considering intrinsic differences and local conditions (e.g., climate and geographical conditions), energy congestions between renewables and hybrid grids, optimisations with advanced algorithms, and multi-criteria decision-making to promote willingness and readiness for stakeholders' participations.

Nomenclature		FHD	following one of the electricity or cooling demand with higher primary energy consumption
Abbreviation		FLD	following one of the electricity or cooling demand with lower primary energy consumption
BIPVs	Building Integrated Photovoltaics	FCD	following the cooling demand method
CFCs	chlorofluorocarbons	FED	following electricity load method
COP	Coefficient of performance	FC	Fuel cell

(*continued on next column*)

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CHP	Combined Heat and Power	GSHP	Ground Source Heat Pump
CFCs	chlorofluorocarbons	HCFCs	hydrochlorofluorocarbons
DES	distributed energy systems	ML	Machine Learning
DCS	district cooling systems	NPV	Net Present Value
DHC	District heating and cooling	RTP	real-time price
DC	District cooing	STC	Solar Thermal Collector
DH	District heating	TOU	time-of-use price

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1. Introduction

The decarbonization with renewable and sustainable developments of district energy systems has been regarded as a critical target for the modern society and natural ecosystems [1,2]. Low-carbon smart cities [3] with advanced simulation tools [4,5] are significant for carbon-neutral transition [6] and energy resilience enhancement [7,8]. Fig. 1 demonstrates the evolution of district energy networks from the

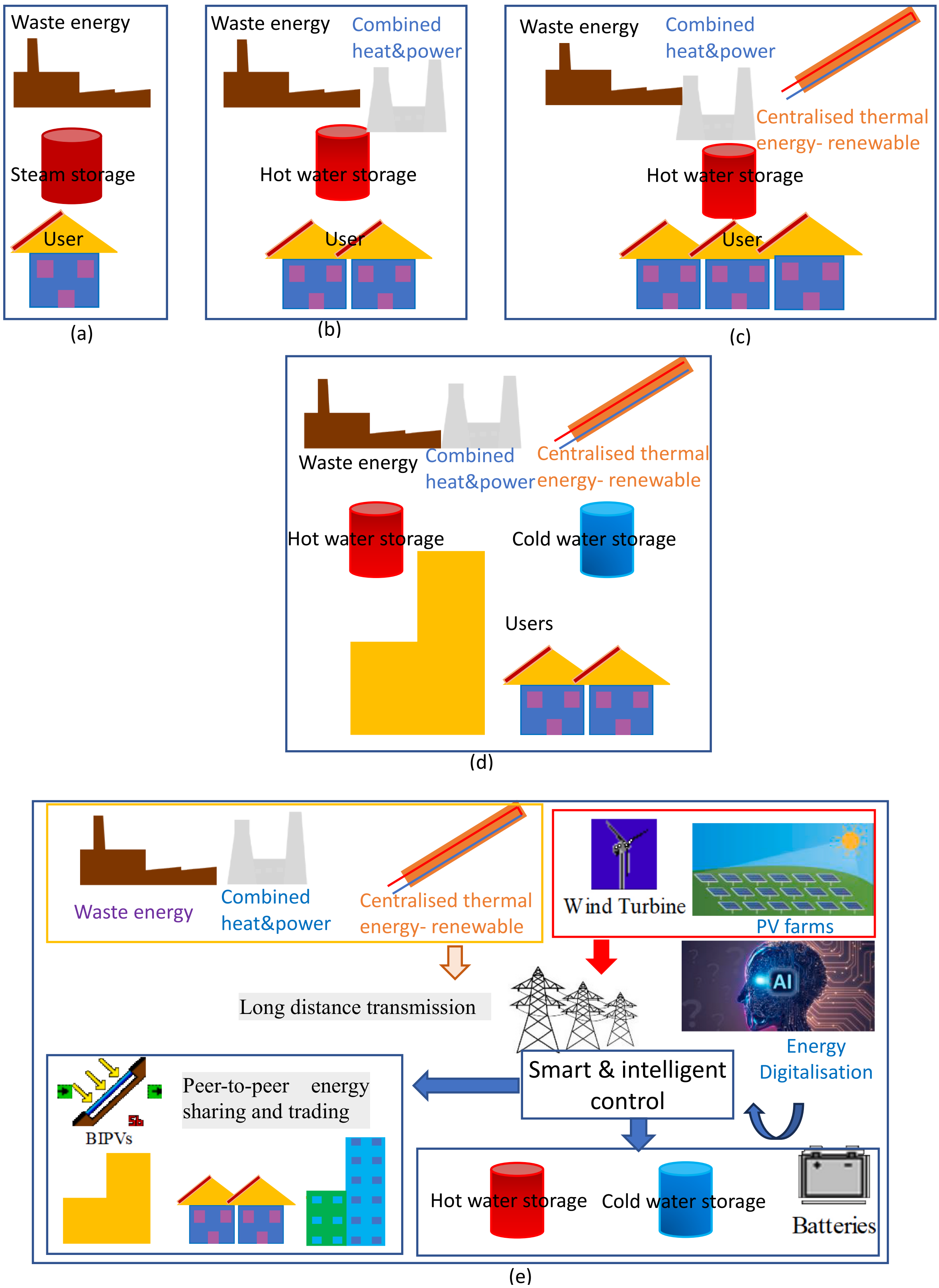


Fig. 1. Demonstration of 5th district energy networks: (a) 1st heat recovery; (b) 2nd district heating; (c) 3rd district heating; (d) 4th district heating and cooling; (e) future digital energy networks.

1st district heating with heat recovery to future digital energy networks. Compared to the 1st district heating, the 2nd district heating integrates the combined heat and power units. Centralised thermal energy for renewable utilisation has been integrated in the 3rd district heating. Furthermore, the 4th district and cooling integrate both cooling and heating storage tanks for renewable penetration improvement. Lastly, the future digital energy network will include a lot of new elements, like energy digitalisation, smart & intelligent control, peer-to-peer energy sharing and trading [9–11], etc. Two concepts related to expanded energy boundaries can be noticed along the evolution of district energy networks from the very early district heating (DH) system to the latest generation, i.e., the increased consideration in waste heat recovery and renewable penetration [1], and more active participation of end-users [1]. District energy systems [12,13] have attracted researchers' interest from perspectives of modelling approaches [14], clean energy production and use [15,16], advanced energy conversions [17,18], high-efficient energy storages [19], and hybrid grids' interactions [20,21]. Over the past several decades, studies have been mainly on district energy systems in different climates. However, a holistic review is missing in combining DH/district cooling (DC) networks, hybrid renewables' integration, diversified energy storages with multiple energy forms and advanced energy conversions, and hybrid grids' interactions. Furthermore, rational analysis on multi-diversified energy systems and associated optimisation methodology is necessary but rare in academia, especially for smart district energy system, in terms of practical feasibility, multivariable and multi-objective optimisations, and the trade-off between computational complexity and optimisation accuracy. Environmental, Science and Government (ESG) and Sustainable Development Goals (SDGs) require combined efforts from countries/regions, research, industry, and policymakers. Di Silvestre et al. [22] explored how decarbonization, digitalisation and decentralization change the key power infrastructures, involving technological evolution, digital revolution, distribution levels and architectures. Inayat and District cooling system with renewable is highly dependent on local government policy [23]. Zhou [24] comprehensively reviewed carbon neutrality transition with advanced energy policies. Results indicate the global policies played pioneer roles for sustainability transition.

Over the past several decades, the aggravation in the energy shortage crisis and the daily increased energy demand leads to a stricter requirement on energy consumption. In response, different novel energy systems from district levels and techno-economic feasibility in different climates have been proposed, studied and investigated. A state-of-the-art literature review on renewable systems with hybrid energy storage for district heating and cooling (DHC) is critical, and potential challenges need to be outlined to promote the widespread application in the future. Energy storage systems are diversified, including thermal storage systems (sensible, latent, and thermochemical storages) and electricity storage systems (such as electrochemical battery, compressed air energy storage, supercapacitor, fly-wheel and hydrogen system). Advanced energy conversions include solar-to-heating, solar-to-cooling, solar-to-electricity, and heating-to-cooling conversions. The hybrid grids' interactions include DHC grids, power grid, natural gas grid, biofuel grid and hydrogen grid.

The alleviation of fossil fuel depletion, green-house gas (GHG) emission reduction, and energy system stability and efficiency enhancement require renewable penetration, efficient energy storage, and smart grids with optimal design and robust control. In respect to district energy systems, multi-criteria include system efficiency [25], electric power saving [26], primary energy consumption [27], equivalent carbon dioxide emissions [28,29], economic considerations (including the investment cost [30] and operational cost [27]) and life-cycle analysis [28]. The straightforward multivariables related to these objectives involve both design and operating variables. To improve energy performance, researchers have conducted both parametric analysis and optimisation. In respect to the parametric analysis, effective strategies were explored to improve energy performance, such as the thermal storage in district heating [26], optimal configurations of power plants based on decentralized heating systems [27], and cloud-based demand-response strategies for DH networks [30]. In respect to system optimisation, adopted optimisation algorithms include linear programming [29], genetic algorithms [31], and hybrid mixed-integer linear with a three-stage heuristic approach [32]. One of the critical challenges in optimisation is the computational complexity, which requires the time-consuming computation process with thousands of threads to identify optimal solutions. The technical feasibility of machine learning methods on the enhancement of optimisation has been reviewed. In addition to optimisations, prospects of machine learning techniques have also been demonstrated, in terms of prediction of district energy demand [33] and robust model development [34,35], surrogate model development for uncertainty and optimisation analyses [36], optimal geometrical and operating parameters, and efficient and robust energy management and control strategy [37,38].

Based on the above literature review, research gaps can be noticed.

1) Optimal planning, design and operation of multi-energy systems require computational complexity, which requires the time-consuming computation process with thousands of threads to identify optimal solutions;
2) Strategies for energy resilience and flexibility enhancement of district heating and cooling energy networks have not been systematically reviewed, in terms of heating networks' design, the demand and grid-responsive control, and the flexible management of hybrid energy storages in accordance with hybrid grids' requirements;
3) Roles of machine learning techniques in energy digitalisation have not been characterised. Functions of machine learning in complicated multi-energy systems are unclear.

Research novelty and originality of this study are summarized below.

1) An interactive energy network was formulated in this study to provide frontier guidance to promote resilient and smart district energy networks in a future smart city, which systematically integrated onsite renewable generations, waste heat recovery from centralised power plants, multi-diversified energy storage, advanced energy conversions for energy sharing;
2) Strategies and effective solutions have been proposed for energy resilience and flexibility enhancement of district heating and cooling energy networks, including heating networks' design, the demand and grid-responsive control, and the flexible management of hybrid energy storages in accordance with hybrid grids' requirements;
3) Functions and roles of machine learning in complicated multi-energy systems have been systematically reviewed, in terms of prediction of district energy demand and robust model development, surrogate model development for uncertainty and optimisation analyses, optimal geometrical and operating parameters, and efficient and robust energy management and control strategy.

2. Methodology

A holistic literature review of district energy systems was conducted following the roadmap, as listed in Fig. 2. Both district heating and cooling systems are involved with hybrid renewable integration and hybrid grids' interactions (such as district cooling/heating grids, power grid, natural gas grid, biofuel grid and hydrogen grid). Five grand thermal challenges were identified (e.g., thermochemical/physical thermal storage [39,40], industrial decarbonization, heat pumping systems, long-distance transmission of heat and variable conductance building envelopes), together with potential solutions [12]. Roles of hybrid renewables, novel energy storage systems, and diversified grids' interactions have been analysed. Afterward, smart district energy systems have been reviewed, regarding optimal design, robust operation, energy controls and resilient grids' interactions. Optimizations on

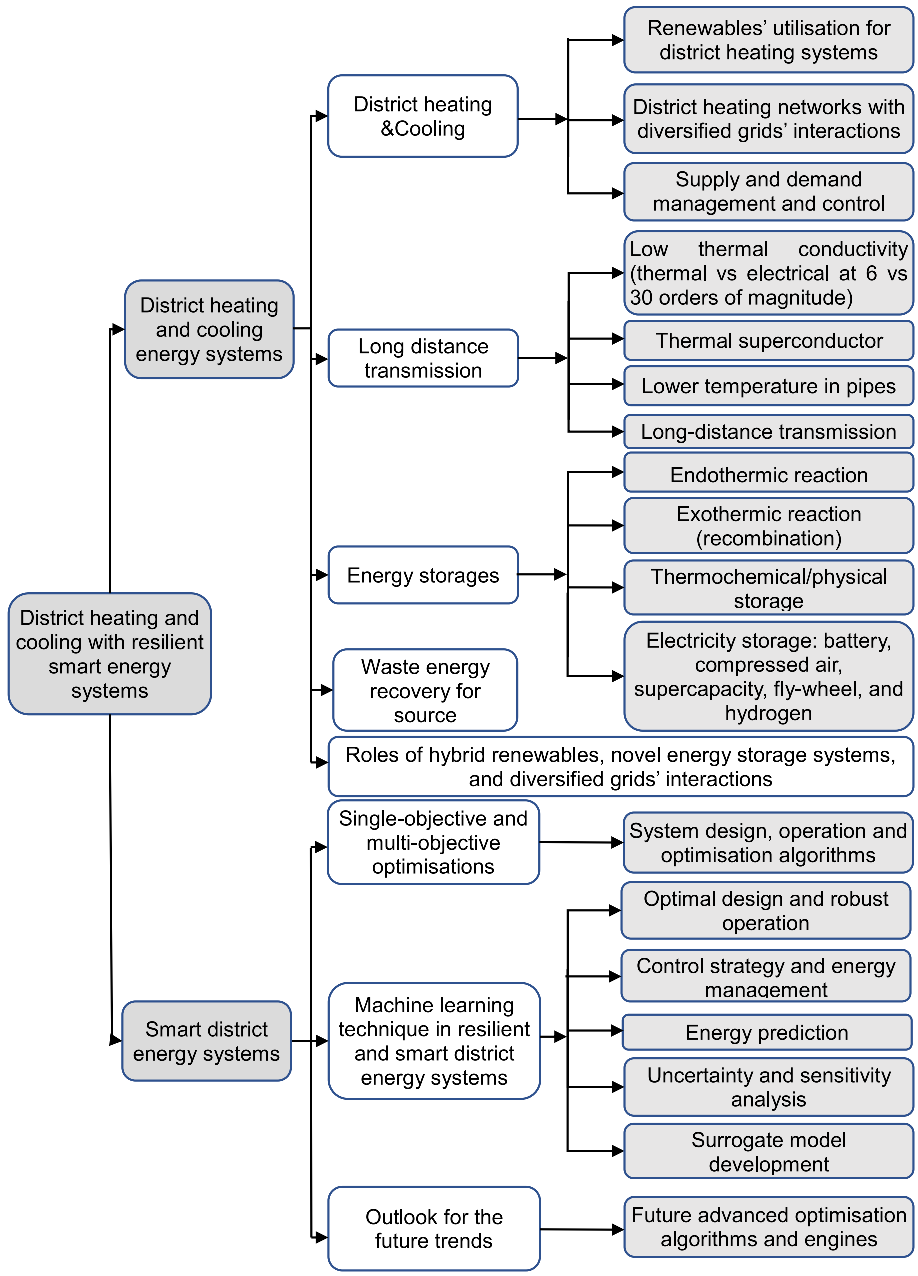


Fig. 2. The roadmap of renewable systems with hybrid energy storages for district heating and cooling.

nonlinear district energy systems were comprehensively presented for the promotion of smart district energy systems. Furthermore, applications of machine learning methods for the realisation towards resilient and smart district energy systems have been analysed, in terms of prediction on district energy demand and robust model development, surrogate model development for uncertainty and optimisation analyses, geometrical and operating parameters' optimal design, efficient and robust energy management and control strategy. Last but not the least, a district energy network has been formulated for the guidance of resilient and smart district energy networks in the future smart city, involving with on-site renewable generations, the waste heat recovery from centralised power plants, multi-diversified energy storages, advanced energy conversions for interactive energy sharing.

In terms of systematic literature selection, the bibliometric approach is adopted by typing keywords in Google scholar. Keywords mainly include district heating and cooling, hybrid energy storage systems, multi-objective optimisations, machine learning, resilient and smart grids' interactions, and etc. The most impactful articles (e.g., high citation) within latest 5 years are given high priority for the selection. Furthermore, considering the globalisation and diversity of the research, priority is also given to the publications in different continents and climates.

This review includes three sections. The first section systematically reviews district energy systems with renewables and hybrid energy storages, including five subsections, i.e., district heating networks, district cooling networks with energy management and control strategy, energy storages with hybrid grids' interactions, roles of hybrid renewables, novel energy storage systems, and diversified grids' interactions, and an in-depth rational analysis. The second section presents smart district energy systems with energy controls and resilient grids' interactions, including optimisations, artificial intelligence in resilient smart district energy systems, and prospects. In the third section, a highly interactive district energy system was formulated to guide flexible district energy networks in the future smart city, involving on-site renewable generations, waste heat recovery from centralised power plants, multi-diversified energy storages, advanced energy conversions for energy sharing. Challenges and opportunities are identified from benchmarks to select the most appropriate energy storage. It considers the difference of mechanisms and local conditions (such as climate and geographical conditions), conflictions of hybrid grids, multivariable and multi-objective optimisations, and multi-criteria decision-making to promote the willingness and readiness for stakeholders' participations.

3. Systematic literature review of district energy systems with renewables and hybrid energy storages

A holistic overview of district energy systems and research objectives are listed in Table 1. From the perspective of building types, the district energy systems include residential buildings, office buildings, commercial/industrial buildings, university buildings, standalone retail buildings, and building clusters. Energy storages in district energy systems include the sensible, latent, and thermochemical/physical storages for thermal energy storage, the electrochemical battery [41], compressed air energy storage, supercapacitor, fly-wheel, and hydrogen for electricity storage. The involved district microgrids include electrical, thermal, and natural gas grids. District energy management and control can be from demand side management, optimisation-PV power predictions and electricity demand, increase in renewable penetrations with integrated energy storage. Multi-objectives are included in the district energy systems, including system efficiency, electric power saving, primary energy consumption, equivalent CO_2 emissions, economic considerations (initial investment and operational cost), and life-cycle analysis.

3.1. District heating networks, energy management and control strategy

3.1.1. Systematic literature review of renewable energy utilisation for district heating systems

In the context of DH, renewable energy utilisation can be classified into heating sources, system formulation, energy conversions and hybrid microgrids' interactions. The energy scales cover from single buildings, office building clusters [92], district buildings [14], cities until different regions [93]. Due to the high efficiency and low cost of the solar thermal system, the parabolic collector has been widely studied in adaption to different climates to maximize thermal efficiency. Table 2 lists a holistic overview of district heating systems, including diversified energy sources, energy storages, advanced conversions and hybrid microgrids' interactions. Researchers mainly focused on the novel design of non-tracking concentrating collectors to improve thermal efficiency [94]. Compared to the commercialized non-tracking collector, the new collector can increase the thermal efficiency by 20 %. The solar parabolic-trough collector using rotatable axis tracking shows more competitiveness due to the real-time vertical incidence of solar radiation to the collectors' center [95,96]. Researchers have paid particular attention to the intermittence and instability of solar energy. On the other hand, geothermal energy is more stable, providing almost constant temperatures by the ground with considerable storage capacity. The energy utilisation forms of geothermal energy for DH include tunnel ventilation [73], drilled boreholes-based ground source heat pumps [53, 97], and geothermal assisted air-conditioning systems [93]. In addition to solar and geothermal energies, the combined heat and power (CHP) was regarded as the most reliable energy source for district heating systems. According to different driving sources driving, combined heat and power systems (CHPs) can be classified into the coal-fired CHP plants [98,99], the biomass-driven combined heat and power [100,101] and the waste-to-energy CHP plant [102]. Compared to the coal-fired or natural gas-fired CHP plants, the biomass-driven CHP and the waste incineration CHP plant are more promising. The transportation networks can contribute to the local district heating grids with the implementation of heat recovery. Davies et al. [103] upgraded the recovered heat from the underground railways and contributed to the local district heating network. Along with the implementation of the proposed heat recovery technique, 50 % of carbon savings can be realized. Revesz et al. [104] recovered the heat from underground infrastructure for efficiency enhancement. The research results indicated that the heat extraction rates of ground source heat pumps (GSHPs), which were close to

Table 1
A holistic overview of district energy systems and research objectives.

Building Type	Residential buildings [42, 43]	Office building [44,45]		Commercial [46]/industrial [47]/office building [48]	University building [49]	Standalone retail building [46, 50]	Residential [51] and commercial building clusters [52]	
District Energy Storages	Thermal energy storages			Electric energy storages				
	Sensible energy storages: ground [53]and molten salt energy storage [51,54, 55]	Latent energy storages: PCMs energy storage [56, 57]	Thermochemical/ physical storage [58,59]	Battery storage [60]	Supercapacitor [61–63]	Fly-wheel [64]	Compressed air energy storage [65–67]	Hydrogen energy storage [68–70]
Renewable Energy Sources	Solar-to-electricity Solar PV	Solar Tower Molten Salt Photothermal Power Generation [71]		Wind-to-electricity Wind turbine [72]	Geothermal energy Earth-to-air heat exchanger [73]	Ground source heat pump [53]	Biomass Bio-mass driven micro combined heat and power unit [74,75]	
District Microgrids	Electricity	Electric grid [76,77]						
	Heating/ Cooling	Electrical + thermal grid [78]						
	Fuel	Natural gas grid [77]						
District Energy Management and Control	Demand side management strategies [50] (e.g., adjustion of the cooling [79], demand response [44], grid response [80], Renewable-to-thermal storages [81,82], and occupants' behaviors [14])			PV power and electrical demand [33], model predictive control (MPC) [43]	Renewable penetrations in buildings with extended energy storage boundary: PEVs [83] and HVs [84]	Participation of end-users [85]		
Objectives	Grid interaction [86,87]	Matching capability [45,88]		Equivalent CO_2 emission [89]	Operational cost [50, 87]	Life cycle cost or net present value [89]	Thermal comfort, power consumption [90]	Energy flexibility [91]

Table 2
The holistic overview of the district heating systems, in respect to diversified energy sources, energy storages, advanced conversions and hybrid microgrids' interactions.

Heating sources		Systems	Energy forms	Storages	Energy conversions	Microgrids' interactions	Scale
Solar thermal energy		Solar thermal collector [105]	Solar thermal energy	Sensible energy storage [106,107]	Solar-to-thermal [105]	Heating grid [108]	Single buildings, office building clusters [92], district buildings [14], cities, regions [109]
		Solar assisted ground source heat pump [110]					
Geothermal energy		Tunnel ventilation [73];	Geothermal energy	PCMs latent energy storage [107]	Power-to-thermal [100,101]	Power grid [107], natural gas [111], waste heat [112–114] and hydrogen fuel grid [115]	
		Ground source heat pump [53,97];					
		Geothermal assisted air-conditioning system [93];					
		Advanced piping system [116]					
Coal/natural gas/incineration		Gas-fired Boiler [117]	Coal/electricity/natural gas/		Bioenergy-to-thermal [101,118]		
		Coal-fired CHP plants [98,99]					
		Waste-to-energy CHP plant [102]					
Biomass		Biomass co-firing [118]	Biomass				
		Biomass driven combined heat and power [100,101]					
Nuclear energy		Nuclear Battery [119]; nuclear district heating [120–122]					
Data center		Low-grade waste heat recovery [123–125]					
Heat recovery	Exhaust heat recovery	Industrial waste heat recovery in district heating [113]; waste-to-energy CHP plant [102]; cascade recovery of flue gas waste heat [114]	–	Thermochemical energy storage [1,126]			
		Engine waste-heat recovery [127,128]	–				
	Heat recovery from transportations	Underground railways heat recovery for district heating network [103,104]	–				

underground railways, could be increased by around 43 %.

As critical and effective breakthroughs for the mismatch between energy generation and energy demand and the load shifting for cost-saving purposes, energy conversion and storage have attracted researchers' interest worldwide. To meet the heating demand, energy conversion techniques include the solar-to-thermal [129,130], power-to-thermal [131], and bioenergy-to-thermal conversions [132,118]. Thermal energy storages include sensible energy storage [133,134], latent energy storage with phase change materials [106,107], and thermochemical/physical energy [126]. The systematic integration of advanced conversions and storages in district heating systems can make fast response and sufficient reaction to stochastic demands by avoiding excess energy production, increasing stability of energy networks, and minimising energy congestions.

Regarding the district heating systems, several technical challenges need to be clarified and investigated in the upcoming studies. Firstly, the rotatable axis in the solar parabolic-trough collector for the solar tracking will suffer from performance degradation over thousands of rotating cycles, and the maintenance cost of the rotatable axis needs to be well considered. Secondly, in respect to the solar-assisted or geothermal-assisted GSHPs, technical challenges remain for the high initial cost of drilled boreholes and the unbalance of the annual heat rejection and extraction in the ground. Novel strategies are necessary for the energy balance in different climates. Thirdly, capacity expansion of microgrids and the development of standardized auxiliary facilities will increase the economic investment during the facility upgrade.

3.1.2. District heating networks with diversified grids' interactions

Synergistic complementarity with combined operation on each integrated system, is required to form a smart energy network with hybrid grids' interactions. The sophisticated building energy systems can acquire the ability to manage renewable generations and energy demands, regarding climate conditions, occupants' demands and local grids' requirements. By dynamically managing renewables, demands, storage [135], and grids' interactions, a smart energy system can cover energy demand with high energy flexibility [136], low CO_2 emission, and reduced reliance on grids. An interactive districting system with renewable, storage, diversified grids' interactions has been formulated, as shown in Fig. 3. Thermal energy sources include solar thermal collectors, geothermal energy and recovered thermal energy from plants and industries. Electricity sources include gasoline-based diesel engines, building integrated photovoltaics (BIPVs), and wind turbine. As a promising hybrid energy generation device, the CHP can provide multilevel supply energy for cascade utilisation. Energy sources driving the CHP include bio-fuel, biomass, and waste incineration. The surplus electricity can be stored in hybrid electricity storages, together with heating energy recovery for domestic usage, such as heat recovery from the fuel cell-based hydrogen storage [137] and friction heat recovery from the compressor. The surplus or recovered thermal energy can be stored in daily thermal storage tank or seasonal thermal storage ground [138]. Hybrid grids dynamically balance energy in the formulated energy system via power grid, district heating grid, biofuel grid, natural gas grid, and hydrogen fuel grid [139].

3.1.3. District heating networks, energy management and control strategy

In order to improve the resilience and flexibility of DH networks, technical solutions can be focused on the design of heating networks, demand-responsive control, and management of energy storages under hybrid grids requirements. In addition to the main participants of building sectors, participants of transportation sectors can perform well in the whole system. The main principle is that, mobile thermal storage systems with vehicles can realise the energy sharing and establish the interactive energy network between diversified heating sources and end-users [138]. For instance, as shown in Fig. 4, trucks can carry goods to the factory. It can also take thermal energy, with a thermal energy storage with high energy density, from the exhaust heat in factory back to cover the heating demand of end-users. In such a scenario, the exhaust heating energy exported to the heating grid from the factory can be reduced and the heating energy imported from heating grid can also be reduced to cover the end-user's heating demand.

From the perspective of heating networks' topology design, the branched pipe network and the annular pipe network can be noticed. In respect to the district heating network with annular pipes, as shown in Fig. 4(b), mobile vehicles with energy storages can systematically connect different buildings, centralized energy storage, exhaust heat sources (like factories), and energy generators (like CHP plants). In addition to mobile vehicles, centralized energy storage systems can contribute to the energy sharing between energy generation and demand units when implementing annular pipe networks. Compared to the sophisticated piping system in branched pipe networks, as shown in Fig. 4(a), energy sharing between different energy generation and demand units becomes more straightforward in annular pipe networks. Along with the participation of mobility and thermal storages, the district heating network with annular pipes, integrating peer-to-peer energy sharing and trading [141], vehicle-to-building interaction, and so on, can be more reliable and flexible than the branched pipe network in terms of power supply reliability, system resilience under extreme and unexpected disruption, and mitigation on grid reliance.

For efficient energy management in the building energy system, researchers have studied demand-responsive control, grid-responsive control strategy, and management of hybrid energy storage. In respect to demand-responsive control, Mishra et al. [142] conducted field testing to study the impact of inlet water temperature-based demand response on indoor thermal comfort. Research results indicate that the demand response will not affect the occupants' perception of indoor thermal environments, while it can provide significant avenues of energy flexibility through thermal mass in the building envelope. Guelpa et al. [143] investigated the demand side management for the construction of optimal control of DH networks, from perspectives of time reschedule of heating systems and modification of parameter settings. They conclude that the peak reduction can be 35 % on the peak request. Guelpa et al. [144] studied demand-side control strategy with different return temperatures (DRT). According to the results, using the DRT regulation strategy can reduce building thermal peak demand by 15 % and peak power by 24 %. In respect to the grid-responsive control strategy, Wang et al. [145] provided a systematic literature review on the development of grid-responsive buildings with fast demand responses to smart grids. Zhou et al. [146] developed an advanced grid-responsive control to provide techno-economic viability and improve energy flexibility to power grid. In respect to the flexible

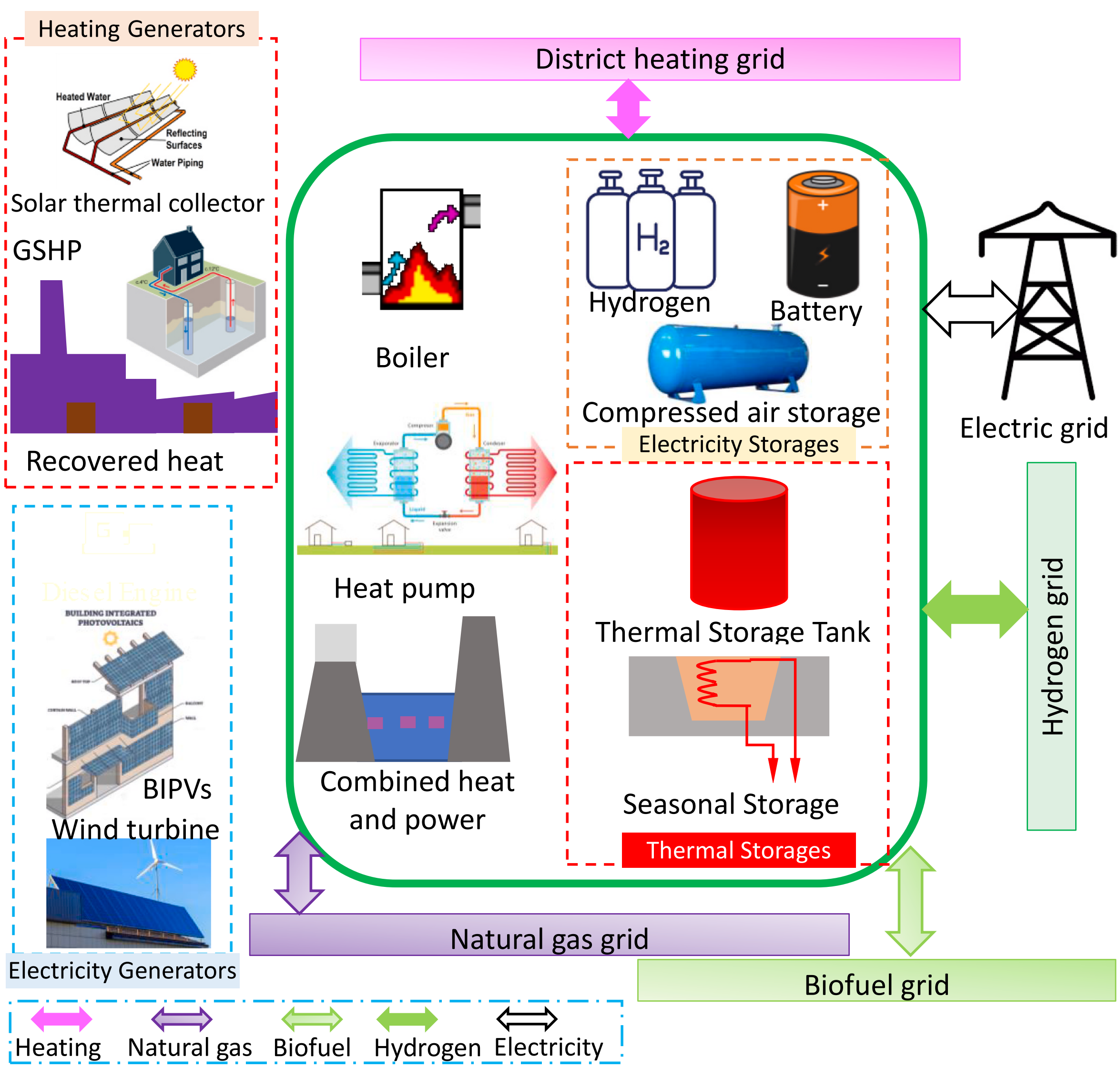


Fig. 3. Renewable-to-electric and thermal storages with multiple energy grids' interactions..
(Note: GSHP refers to ground source heat pump.)

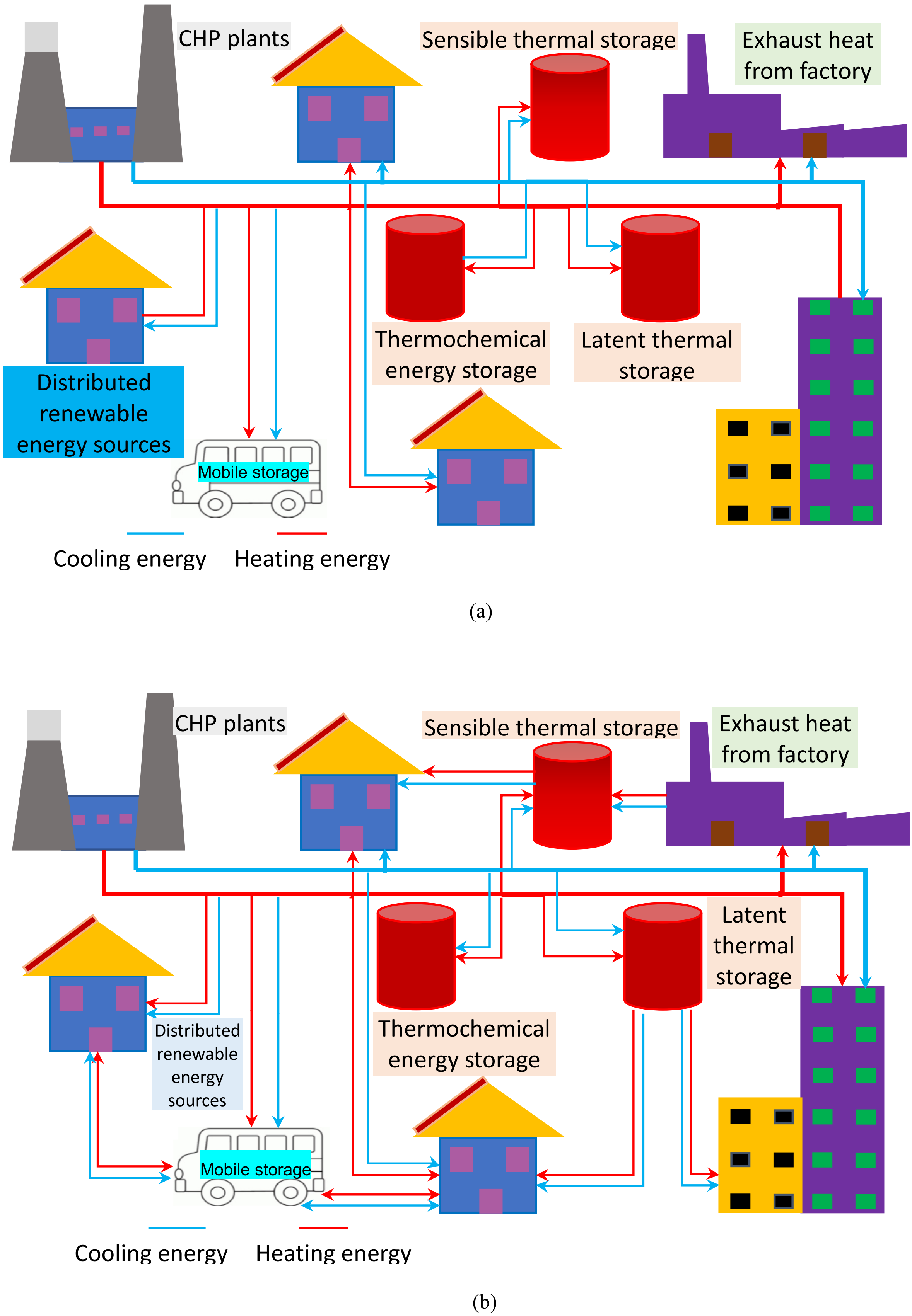


Fig. 4. TES connected to district heating network, in respect to (a) branched pipe networks [140]; (b) annular pipe networks [140].

Table 3
A holistic overview of the district cooling systems, in respect to diversified energy sources, energy storages, advanced conversions and hybrid microgrids' interactions.

Cooling sources	Systems	Storages	Energy conversions	Microgrids' interactions	Scale
Solar thermal energy	Absorption cooling system [160,105] Adsorption cooling system [160,105] Desiccant cooling system [178,179]	Sensible energy storage [163,175]	Solar-to-power [151,163]	Smart heating grid [176,177]	Single buildings [149,150], residential building clusters [151], district buildings [152,153], cities, regions [154]
Geothermal energy	NH3–H2O refrigeration [164] Single-stage Li–Br water absorption cycle [165,166] Absorption heat pump [167] Geothermal assisted air-conditioning system [169]	PCMs latent energy storage [180]	Solar-to-Power&cooling [161,162]		
Solar photovoltaics	Electricity-driven compression chillers [149] PV assisted cooling system, solar cooling in envelope [161,162], solar PV air-conditioner, Solar thermoelectric cooling-heating system [163]	Electric storage [163,153]	Power-to-thermal [175,181]	Heating and power grids [78]	
Sea water	seawater shower cooling towers [172] seawater air conditioning [173,174]		Heating-to-cooling [105,181]		
Biomass	Biomass-driven absorption chiller [153] municipal solid waste driven power and cooling [170,154] biomass boiler integrated absorption chiller [175]	Thermochemical energy storage [126]		Power grid	
Coal/Disel/natural gas	Centralized [181] and decentralized CHPs [151,170]		Bioenergy-to-thermal [181]	Heating and fuel grids [182]	
Heat recovery – Exhaust heat recovery	Adsorption cooling with waste heat recovery from internal combustion engine [171]				
Heat recovery – Brake heat recovery	Regenerative braking				

management of hybrid energy storage under hybrid grids' requirements, Lu et al. [147] developed a coordinated dispatch strategy to manage the power in a DH network with battery storage. The proposed coordinated dispatch can effectively improve the flexibility and economic performance of the multi-energy system. Pakere et al. [148] comparatively investigated the techno-economic competitiveness of different energy management strategies. Results showed that excess power-to-heat conversion is the best option when the electricity price is low.

3.2. District cooling networks, energy management and control strategy

3.2.1. Systematic literature review of renewables' utilisation for district cooling systems

In the literature, renewable utilisation in district cooling can be classified into cooling sources, system configurations, diversified energy storages, energy conversions, and hybrid microgrids' interactions. Table 3 provides a holistic overview of district cooling systems. The energy scales covers from single buildings [149,150], residential building clusters [151], district buildings [152,153,105], cities until regions [154]. Traditional vapor compression refrigeration systems employ refrigerants with a high global warming potential. The chlorofluorocarbons (CFCs) and hydrochlorofluorocarbons (HCFCs) gases result in the ozone layer depletion and global warming. Sorption chiller systems can employ solar energy to replace the vapor compression refrigeration system, and thus mitigate the global warming [155,156]. The solar-driven cooling system can replace the vapor compression refrigeration cycle with thermal-driven absorption/adsorption chillers. Within the solar thermal driven district cooling system, the solar thermal collectors are connected with thermally driven chillers to generate cooling energy for buildings [157,158]. Hassan et al. [159] investigated and validated the technical feasibility to drive the adsorption cooling chiller for cooling energy generation, with low-grade solar thermal energy. Allouhi et al. [160] provided a comprehensive literature review of different solar-driven cooling systems, including absorption system, adsorption system, and solid desiccation cooling system. The pros and cons of each solar driven cooling system were presented, in respect to different climates. Furthermore, solar-to-power-to-cool systems also show promising competitiveness. These systems include PV assisted cooling systems, façade-integrated solar cooling [161,162], solar PV-grid powered air-conditioner, and solar thermoelectric cooling-heating system [163]. However, due to the intermittence of solar energy, these systems are relatively volatile. By contrast, the geothermal-assisted cooling system can potentially provide more reliable and robust performance than the solar driven system, as the geothermal energy is more stable due to the almost constant temperature of the ground with good storage capacity. For the DC purpose, the geothermal energy can be integrated with NH_3–H_2O refrigeration [164], single-stage Li–Br water absorption cycle [165,166], absorption heat pump [167], adsorption cooling [168] and geothermal assisted air-conditioning system [169]. In addition to solar and geothermal energies, the CHP system has been regarded as the most reliable energy source for DC systems. Energy sources of CHPs can be biomass [153], municipal solid waste [170,154], coal, diesel and natural gas. The waste heat recovery technique can be integrated into the DC system. Hamdy et al. [171] provided a systematic literature review on adsorption cooling systems to recover waste heat from an internal combustion engine. The adsorption cooling system can improve the overall engine efficiency and reduce carbon dioxide emissions. However, the low performance of the waste heat-driven system is the main bottleneck for its wide acceptability in district cooling systems. Furthermore, in the coastal regions, seawater can be a promising source for district cooling, such as seawater shower cooling towers [172] and seawater air conditioning systems [173,174]. However, the corrosion of piping systems from salty seawater is the main barrier to its widespread applications.

Energy conversions and thermal energy storages are effective to improve energy performance of renewable supported district cooling

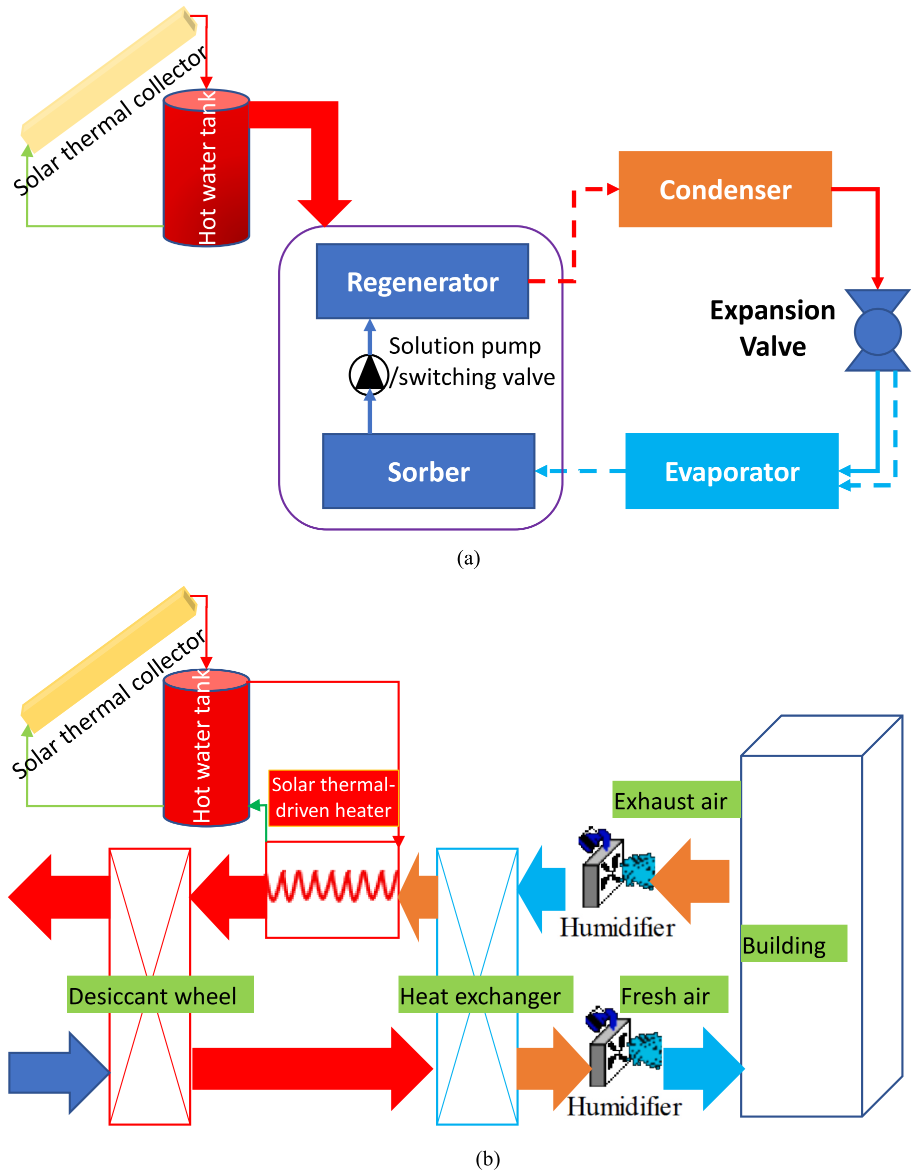


Fig. 5. The mechanism of solar thermal-driven cooling systems: (a) sorption system; (b) desiccant cooling system.

system. The adopted energy conversion techniques include solar-to-power [163], solar-to-power and cooling [161,162], power-to-thermal [175,183], heating-to-cooling [105,181] and bioenergy-to-thermal [181]. Energy storage have been widely applied to enhance the renewable penetration and demand coverage concerning the use of surplus energy from renewables and energy conversion devices. Energy storages for district cooling can be classified into sensible energy storage [163,175], PCMs latent energy storage [180], thermochemical/physical energy storage [126] and electric storage [163,153]. Effective management in DC can make fast response and reaction to cooling demands with an enhancement of stability and resilience of energy networks.

Regarding the DH systems, several technical challenges need to be clarified and investigated in upcoming studies. Firstly, the driving temperature for conventional adsorption cooling systems is often higher than 80 °C [184], requiring research efforts on new sorbent materials to lower the regeneration temperature [185,186]. Secondly, in solar-to-power-to cool system, the heat transfer fluid, CFCs, in the refrigeration cycle process will result in depletion of the ozone layer and global warming. Although adopting an absorption or adsorption cooling system to replace the vapor compressor can avoid the use of freon, the low efficiency of the sorption cooling systems is the main bottleneck for its widespread application. Thirdly, there might be less resources (e.g., initial investment cost due to limited budget and limited space) for the district cooling grid from the district multi-energy level, especially with intensified congestion among multi-diversified energy grids. The activations of synergistic functions (such as the complementary battery and hydrogen energy systems under idling mode [187,188]), considering differences of hybrid grids' requirement with diversified energy conversions, are practical solutions to improve the resilience and flexibility of multi-energy systems.

3.2.2. District cooling networks with diversified grids' interactions

Regarding the cooling energy sources for DC, solar-driven cooling systems account for a significant share in all renewable-driven cooling systems. The solar resource is abundant in cooling-dominated regions, such as a subtropical region, Hong Kong. Generally, solar-driven cooling systems include solar-to-power-to-cool systems and solar-to-thermal-to-cool systems. The solar-to-thermal-to-cool system is more environmental-friendly, more resilient to weather changes and potentially less expensive.

As shown in Fig. 5(a), the main principle for the replacement of the compressor is described below. Solar thermal energy is the main driving

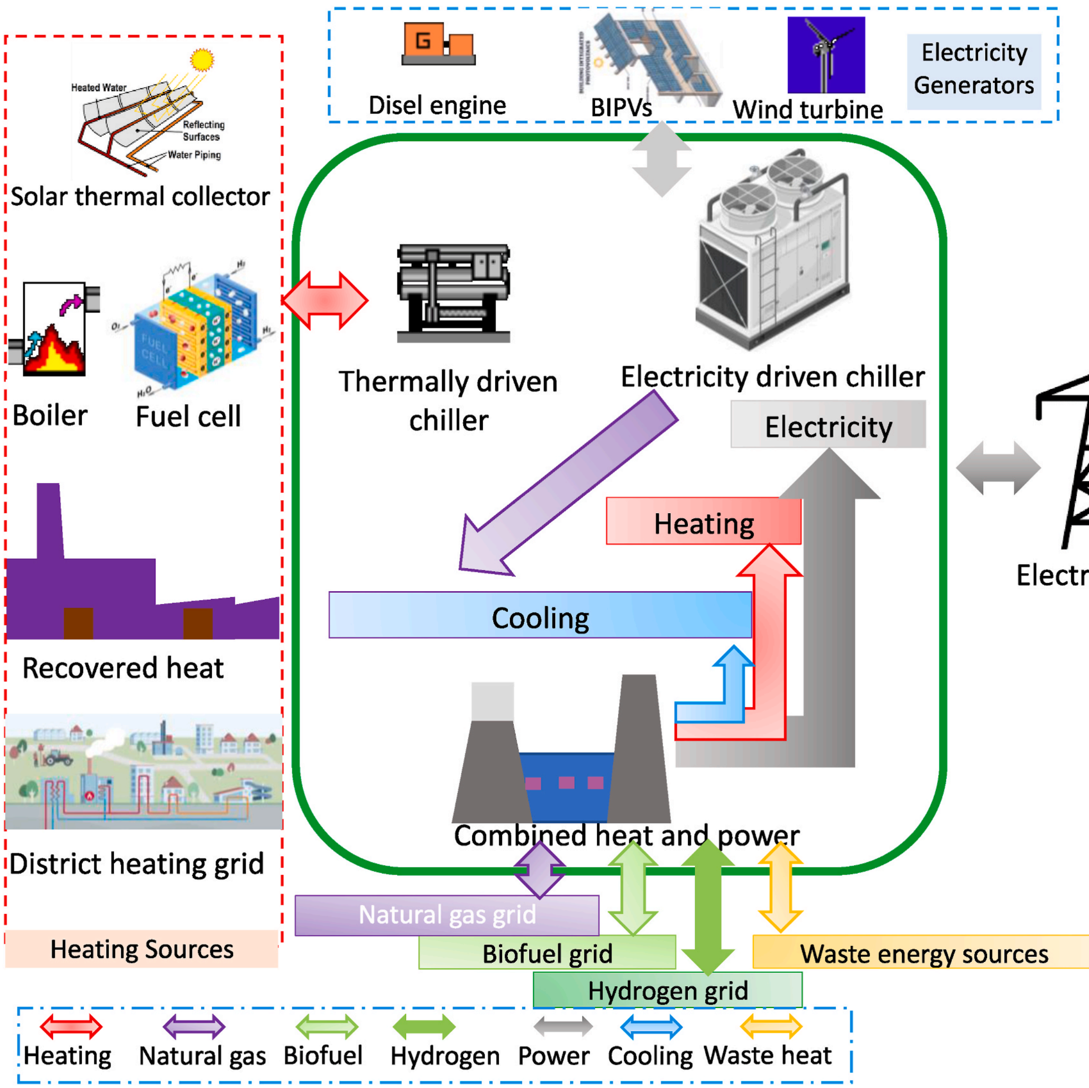


Fig. 6. District cooling networks with diversified grids' interactions.

force to separate the refrigerants from the dilute solution or the adsorbent. After being released, the refrigerants will enter into the condenser, and the concentrated solution or the adsorbent is ready to drive refrigerant and create cooling effect in a new sorption cycle. Fig. 5(b) shows the principle of desiccant cooling. The hot and humid air enters and is thereafter dehumidified by a desiccant component (coated heat exchanger, wheel, etc.) Meanwhile, the air is heated by the adsorption reaction. Afterward, the supply air goes through a heat exchanger for cooling energy recovery from a cold source (a cooling tower, an evaporator, the exhaust air, etc.) The supply air was cooled and humidified before flowing into the room, if evaporative cooling is used. The humidifier cools the exhaust air from the room and transfers cold energy to the supply air in the heat exchanger. Afterward, the air is reheated by a solar heater, and then transfers heat to the inlet air in the desiccant component. The primary function of solar energy is to heat the exhaust air, which is operated to regenerate the desiccant component and pre-

Table 4
A holistic overview of diversified thermal energy storages.

Thermal storage forms	Types	Applications	Advantages	Disadvantages	Technical challenges
Sensible energy storage [133]	Water and aquifer	Water storage tank; Ground source heat pump	High specific heat capacity and small volume for energy storage	Upper temperature limit; liquid leakage problem	The low energy density results in large storage volume
	Rock and bricks	Thermal mass storage	Higher temperature endurance; no leakage problem; good conductivity	low specific heat capacity and large volume for energy storage	
Latent energy storage	Organic	Building envelopes, centralised and decentralized thermal storage and so on	Non-corrosives; no subcooling; Chemically and thermally stable; Adjustable transition zone	Low enthalpy and thermal conductivity; Flammability; Low density	Phase segregation and thermochemical instability over charging/discharging cycles
	Inorganic		High enthalpy; High thermal conductivity; Nonflammable; cheap	Corrosion; Phase segregation; Subcooling; Thermal stability	
Thermochemical/ physical energy storage	Direct and indirect systems	Concentrating solar power	1) higher energy density; 2) negligible storage loss and higher efficiency for integration with solar power systems.	sophisticated system control, service life, and material compatibility issues and chemical reactions	Performance degradation and material compatibility issues

heat the inlet air.

DC networks have been formulated, as shown in Fig. 6, in respect to thermal and electric generations and hybrid grids' interactions. Thermal energy sources include solar thermal collectors, geothermal energy, and recovered thermal energy from fuel cell systems and centralized power plants. The thermal energy is to drive the sorption chiller to cover cooling loads in district networks. Electricity generation from the gasoline-based diesel engine, BIPVs, and wind turbine can drive the compressor-based chiller to cover cooling load. Furthermore, the CHP can provide multilevel energy for cascade utilisation, with sources from natural gas, biofuel, hydrogen, and waste energy sources. The main function of hybrid grids is to balance diversified energies in the formulated energy system dynamically. Several critical research objectives are worthy of being well investigated: 1) the flexible operations of different devices with diversified energy sources, to improve the overall performance [187] and energy resilience to extreme and unexpected disruptions [189]; 2) synergies within hybrid grids for the flexibility enhancement of DC networks.

3.2.3. DC networks, energy management and control strategy

To enhance the resilience and flexibility of district cooling networks, researchers have mainly focused on demand-side management, integration of renewable and heat recovery systems, and flexible energy management and control strategy. Peltokorpi et al. [190] proposed an organizational system for economic implementation of demand-side control in DC, systematically integrating buildings, systems and customer's understanding.

Sameti et al. [191] indicated that more than 67 % of CO_2 emission reduction could be realized through the solar and heat-driven district cooling, together with cost savings, by optimally designing solar-driven DC systems. Integrating distributed renewable energy with DC can save costs and mitigate grid dependence [192]. Advanced control strategies can cover 81.6 % and 82.9 % cooling loads. Hsu et al. [193] utilised waste heat to drive DC and electricity consumption and CO_2 emission can be reduced by 955.8 MWh and 675 tons per year, respectively.

Cox et al. [194] developed a real-time control to reduce operational costs. Compared to the system with a simple scheduled operation, the real time optimal control with model predictive control strategy can contribute to 16 % of average savings for price.

Challenges should be addressed in respect to renewable integrated DC networks with efficient energy management and advanced control strategies. Firstly, comprehensive considerations are needed for demand-side management within building clusters, including diversity of occupants' behaviors, building types, specific requirements of each grid (e.g., power restraint, ramp rate), conflictions between grids, and peak power mitigation strategies. Secondly, flexible energy management and control have to consider not only the requirements of surrounding energy networks but also barriers and motivation of involved stakeholders, i.e., end-users, building owners, facility managers, aggregators, district and transmission operators.

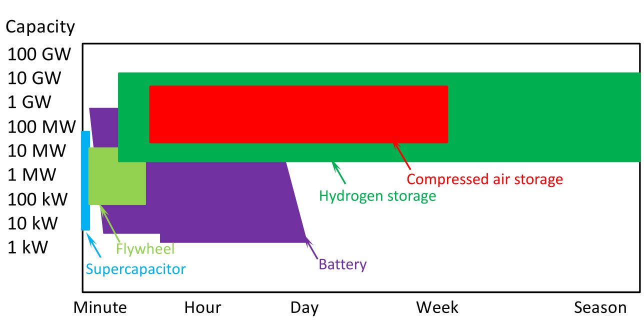


Fig. 7. The storage capacity and time-duration of energy response, with respect to different electric energy storage systems.

3.3. Hybrid energy storages in district energy networks

Thermal storages are necessary to improve the power supply reliability of the energy supply network, the readiness to intermittent thermal demand, and the flexibility through demand-side management in response to requirements of DH grids with multi-diversified sources. Table 4 lists a holistic overview of diversified thermal energy storage. Guelpa et al. [140] reviewed thermal storages in DH, from the pros and cons of thermal storages, such as thermal energy storage forms, key performances indicators, and multi-energy systems. The heating sources include CHP plants, exhaust heat recovery from industries, and distributed renewable energy from on-site buildings integrated renewable systems. Latent thermal storage such as PCMs have been intensively investigated in various applications for the fair storage capacity and good thermos-chemical stability, such as solar thermal storage [195, 196], geothermal energy storage, exhaust heat recovery [197]. Examples also include thermal energy management with electricity-driven devices, such as heat pumps and CHPs. Latent energy storage systems are widely used in distributed energy systems. The thermochemical/-physical storage works with the principle to store energy in chemical form. Compared to latent or sensible heat storage, advantages of thermochemical/physical storage can be briefly summarized: 1) higher energy density due to the endo/exothermic reaction; 2) negligible storage loss and higher efficiency. However, technical challenges resulting from sophisticated system control, service life, material compatibility issues need further studies.

In addition to thermal energy storage, electric energy storage is also critical, due to the high energy quality of electricity. Fig. 7 shows a summary of diversified electric storage systems regarding storage capacity and time-duration of energy response. Electricity energy storage systems mainly include the electrochemical battery, compressed air energy storage, supercapacitor, fly-wheel and hydrogen, energy storage. Compared to the minute-based supercapacitor, the battery storage system shows a broader range of time duration for energy response and a larger storage capacity. Hydrogen storage offers the most extensive range of the time-duration in energy response with an elastic storage capacity. Because water is the only byproduct, hydrogen storage is promising in the DH system in terms of clean power production and climate change mitigation. Table 5 lists a holistic overview of diversified electric energy storage, together with advantages and disadvantages for each system. Several technical challenges should be investigated, including dynamic performance depreciation over long-time operations, operational security, enhancement of energy storage density, and the energy loss during the charging process.

The mobility consumption from transportations can be integrated into the smart district energy system, as the transportation sector consumes 28 % of total energy consumption with 23 % carbon emission. In addition to thermal and electric energy storage systems, the regenerative brake system is a reliable and promising energy source. Fig. 8 shows the cascade energy efficiency from transportations and the hierarchical topography for energy uitilisation. The input energy in the transportation system includes the energy to the engine and the engine loss. For the engine system, a proportion of energy is sent to the braking system, and the rest becomes the rolling and aerodynamic loss. For braking, the first generation for heat management is burning through resistance which is the mainstream, and the second generation is via battery but the grids often reject the feed-in due to the instability. The third generation is supercapacitor. The energy recovery from the braking system, called energy recuperation, is one of the important sources of input energy. Depending on transportation scenarios, different proportions of recovered thermal and electrical energy can be noticed, as shown in Fig. 8(b) and (c), respectively.

Table 5
A holistic overview of diversified electric energy storages.

Electricity storage forms	Types	Applications	Advantages	Disadvantages	Technical challenges
Electrochemical batteries	Flow battery	Load leveling, power quality and emergency standby	Long life cycle, high power, fast working response	Large space, high cost	1) The battery depreciation over thousands of charging/discharging cycles; 2) The environmental pollution from exhausted retired batteries; 3) The security concerns during operational process.
	Sodium-sulfur battery	Load leveling, peak shaving, power quality and renewable energy integration and management	High energy densities power and efficiency	High cost, safety problems	
	Nickel–cadmium battery	Remote control, lamps, aircraft, and diesel engine starters	Low cost	Short life cycle, low capacity	
	Lead-acid battery	Grid storage, multiple deep-cycle lead–acid batteries	Low cost	High self-discharge rate, short life cycle, slow to charge.	
	Lithium-ion battery	Grid storage, regulation and power management services	Long life cycle, high efficiency and energy densities,	Special charging technology, high cost	
Compressed air energy storage	Constant-volume storage	Tank with high-strength carbon-fiber		Pressure level below a certain safety limit	Energy loss in the compressor
	Constant-pressure storage	Storage vessel on the bottom of the chosen water reservoir	Full exploitation of air without safety concerns from Pressure; Flexible locations for geographic positions, such as coastal lines, floating platforms [198];	High cost	
Supercapacitor	Double-layer capacitors Pseudocapacitors Hybrid capacitors	Consumer electronics; Grid power buffer; Low-power equipment power buffer; Voltage stabilizer; Ultra-Battery	Unlimited cycle life; fast charge/discharge cycles for emergency; no full-charge circuit required; Low impedance; Cost effectiveness	Low energy density around 1/5-1/10 of a battery; Low voltage cells; voltage balancing element required; High self-discharge as compared to electrochemical batteries	Energy density enhancement for storage
Fly-wheel	High Velocity Flywheel (angular velocity at 30000–60000 rpm)	Electricity grids for protection against interruptions; wind turbines; motor driven generator to store energy	independent on temperature changes; environmental-friendly with inert or benign materials; lifespan more than 200 years [199]; sealed devices with minimal maintenance; Less thermal loss; More round-trip efficiency; Ease of operation; More peak-load capability	Strong containment vessels for safety precaution	Decrease of energy loss in the friction
	Low Velocity Flywheel (angular velocity up to 10000 rpm)			frictions result in 20 %–50 % of energy loss in 2 h	
Hydrogen energy storage	Low-pressure hydrogen tank	Coverage of electrical and heating demands in district heating networks; transport	Clean energy source, non-toxic	Expensive; safety issue; lower density than gasoline; difficulty in moving with large quantities	Energy density enhancement for storage; Utilisation security
	High-pressure hydrogen tank				

3.4. *Roles of hybrid renewables, novel energy storage systems, and diversified grids' interactions*

Regarding the solar-to-power conversion, the solar tower molten salt photothermal power generation system shows promising prospects in solar-abundant regions with centralized solar collectors. Fig. 9(a) demonstrates the general principle of the solar tower molten salt photothermal power generation system. Compared to photovoltaic systems, the main advantages of the solar tower molten salt photothermal power generation system can be briefly summarized as 1) less dependence on intermittent solar energy with heat storage of molten salt, which makes the 24-h solar-to-electricity conversion possible; 2) the initial and operational costs fall rapidly, due to fast-developed technology and large-scale batch production; 3) the cogenerated heating energy can be more easily recovered, due to relatively mature technologies in heat transfer enhancement. However, the main challenge for the technology is the low efficiency of the thermally-driven electricity generation process due to the energy quality difference between thermal and electrical energy. Fig. 9(b) shows a renewable-fuel cell energy storage with hybrid grids' interactions. The centralised renewable systems can cover the energy demands of building communities and convert the surplus to clean hydrogen. The byproduct, oxygen, can be used to support the life or as fuel for heating energy generation. The cogenerated heat in the proton exchange membrane fuel cell can be recovered to cover the energy demand of the building community. The main advantages for hydrogen storage in district energy applications include 1) the clean energy storage with low environmental pollution; 2) the considerable heat recovery is fuel-saving and reduces emissions for DH applications.

In the district energy system, the main function of diversified grids' interactions is to dynamically balance the hybrid renewables and energy demands. Askeland et al. [202] investigated the impact of DH on the flexibility potential of hydropower resources in Norway. The research results indicate that, by shifting from individual electric heating to DH, the maximum load on hydropower facilities can be decreased, and the hydropower system can dynamically balance the year-round electricity demand. Furthermore, the grids' resilience to extreme and unexpected disruption, and fluctuations in the energy supply, can be improved through advanced energy conversions between different energy grids in district energy systems. In respect to energy management, both demand-side management and energy storage can be operated to manage renewable energy, according to climate conditions, users' needs and grids' requirements. The availability of advanced energy conversions can promote renewable penetration, energy sharing, and grid independence, with complementarity between diversified energy forms. The complementary interaction between energy efficiency and demand response can bring economic benefits.

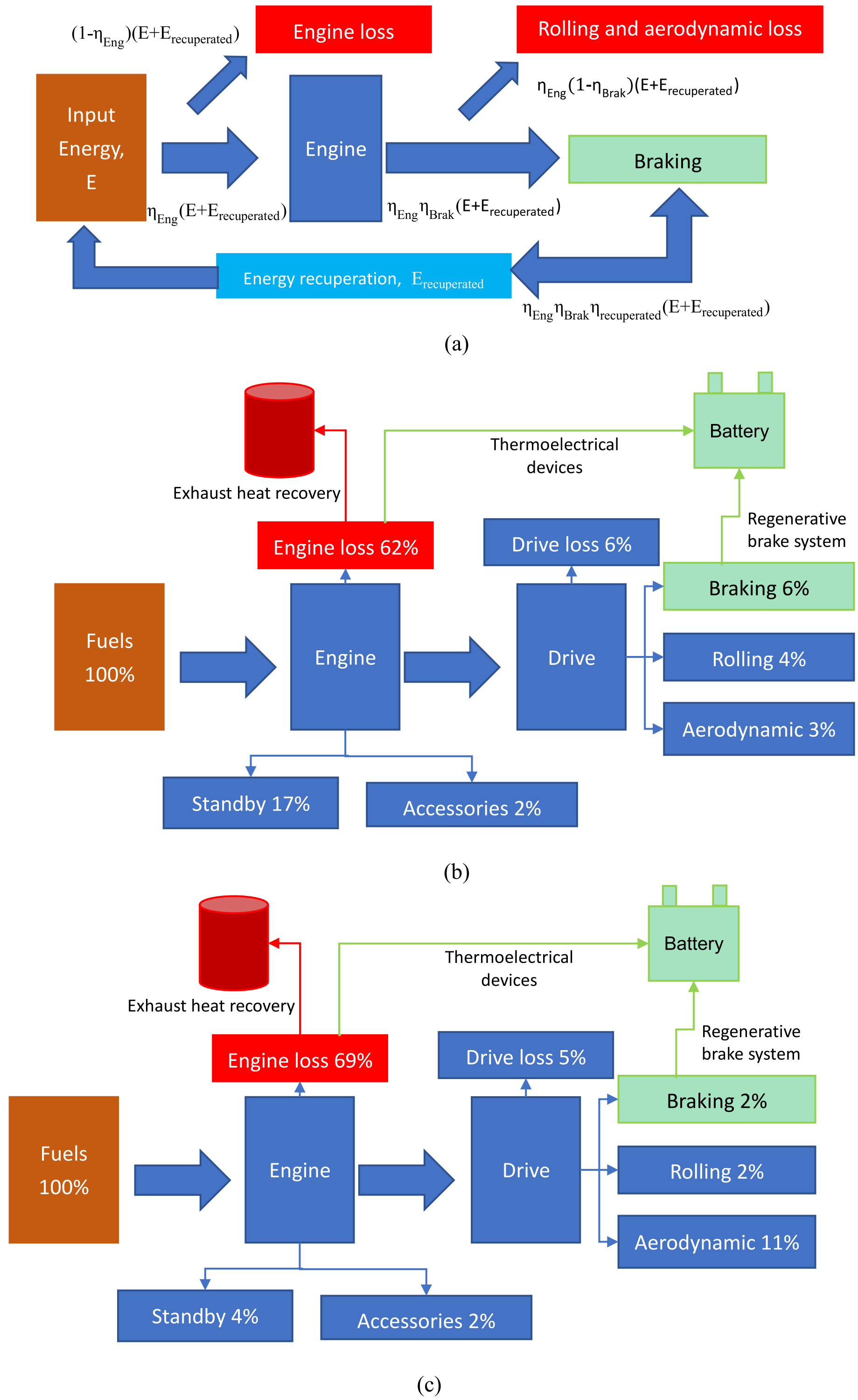
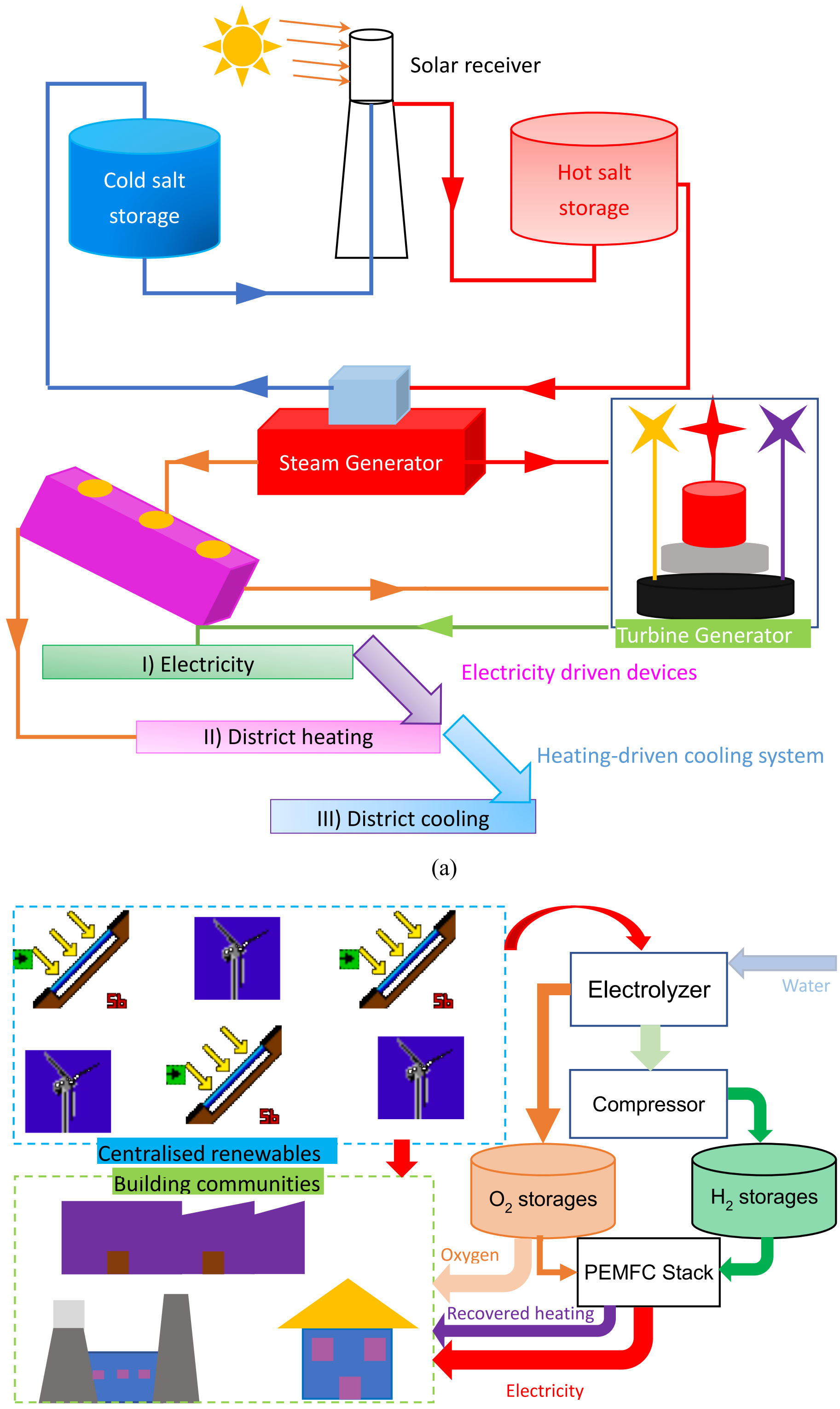


Fig. 8. Cascade energy efficiency of transportations: (a) a hierarchical topography for energy uitilisation; (b) towns; (c) motorways [200,201]. (Note: η_{Eng}, η_{Brak} and $\eta_{recuperated}$ refer to the engine efficiency, brake efficiency, and recuperated efficiency).

3.5. *Applications and future potential*

District energy systems are reviewed on renewables, energy conversions and storages. Both DH and DC systems have been formulated and investigated, from perspectives of renewable-to-electric and thermal storages, energy management, and control strategy for multiple grids' interactions. Mobile vehicles, as mobile energy storages, can systematically connect different buildings, centralized energy storages, exhaust heat sources (like factories), and energy generators (like CHP plants) [203]. Furthermore, grid-interactive buildings can promote load shed intensity and variability. Regarding the formulation of energy networks, the annular piping system is more reliable and flexible than the branched pipe network, in terms of power supply reliability, system resilience under extreme disruption, and mitigation on grids' pressure. However, several critical research challenges can be identified.



(a)

(b)

Fig. 9. (a) The diagram of the solar tower molten salt photothermal power generation system with hybrid grids' interactions; (b) the schematic diagram of renewable-fuel cell energy storage with hybrid grids' interactions.

1) peak power mitigation strategy when enlarging the energy boundary from a single building to building clusters;
2) synergies within hybrid grids to enhance resilience and flexibility of districting energy network;
3) hybrid grids' requirement in demand response and conflictions between different energy grids.

Future studies can focus on.

1) peak power mitigation strategy through combined cooling, heating and electrical energy storages, together with energy dispatch and power control strategies;
2) synergistic operation and compensation among different types of energy sources, in terms of energy density, and demand response time. Furthermore, transition from traditional centralised power system to distributed power typology will promote the energy resilience of the multi-energy system, when suffering from climate change or extreme events;
3) trade-off solutions among different stakeholders and grid utilities for cost-benefit allocations.

4. Smart district energy systems with optimal design, robust operation, energy controls and resilient grids' interactions

Optimisation methodologies of multi-energy systems in district levels were reviewed, demonstrated and discussed, to give an insight into theoretical mechanisms of metaheuristic optimizations. Afterward, trade-off strategies between computational complexity and optimisation accuracy are proposed to promote further development. Moreover, applications and future potential are clarified.

4.1. Optimizations of district energy systems for the system design and operation

In respect to diversified energy systems from the district level, energy systems are complex in design and operation due to multiple parameters and criteria. Generally, in-depth analyses of the sophisticated district energy systems can be classified as renewable and distributed integrations, system design and planning, and system operation and control. Research objectives are multi-dimensional, as listed in Table 6, including system efficiency [25], electric power saving [26], primary energy consumption [27], equivalent carbon dioxide emissions [27–29], economic considerations (including the investment cost [30] and operational cost [27]) and life-cycle analysis [28].

Table 6 lists a holistic overview of optimal design and robust operation of district energy systems in respect to multi-diversified energy systems. Parametric analysis and system optimisation were studied by researchers with respect to single and multi-objectives. In respect to the parametric analysis, researchers focused on practical solutions, such as optimal configurations of a power plant based on a decentralized heating system [27] and cloud-based demand-response strategies for district heating networks [30]. In respect to system optimisation, the adopted optimisation algorithms include linear programming [29], genetic algorithms [31], and hybrid mixed-integer linear with a three-stage heuristic [32]. The optimisation parameters include both design and operating variables. Manente et al. [26] optimized thermal energy storage to reduce the consumption of natural gas. The results indicate that, by charging the thermal storage with a waste incineration plant, the peak demand can be reduced, together with the decrease of natural gas consumption by 10 %. To achieve a resilient and efficient energy transmission network, Li et al. [176] systematically reviewed optimal design on distributed systems. Through the comprehensive literature review, most adopted optimisation algorithms for the optimal system design and reliable operation include mixed-integer linear program, multi-objective evolutionary algorithm, etc. However, due to the complexity of mathematical models through dynamic programming, the optimisation processes are time-consuming and internal memory-consuming, which puts a high requirement on both software and matched hardware, such as supercomputing equipments.

Note that many studies for renewable penetration in DH, transmission loss, and demand-side management have significantly simplified the load patterns in building simulation. To improve the optimisation performance with trade-off strategies between computational efficiency and accuracy, several researchers focused on the development of new optimisation engines, from the perspectives of optimisation mechanisms. Heijde et al. [207] optimized a solar-driven DH energy system using a newly developed multi-objective optimisation toolbox, modesto. The computational efficiency can be improved by 10–30 times without sacrificing the accuracy of optimisation results. Ikeda et al. [208] proposed a new evolution-based constrained optimisation method to optimize district energy systems. Unlike traditional dynamic programming with the searching principle for all samples, the underlying mechanism for the performance enhancement is the stochastic move of individuals to other search points with uniform random numbers, following a classical genetic algorithm. Research results indicated that, the computational load of the proposed method is only 1/457 of the traditional dynamic programming, with the sacrifice of the accuracy only by 2.1 %.

4.2. Machine learning based technique in the resilient and smart district energy systems

Interdisciplinary studies between machine learning and district energy systems have attracted widespread attention in the academic world. Machine learning techniques can be used in the prediction of district energy demand [33] and robust model development [34,35], surrogate model development for uncertainty and optimisation analyses [36], optimal design on geometrical and operating parameters, efficient and robust energy management and control strategy [37]. Moreover, data-driven technique has also been applied in building energy retrofit. Ahmad et al. [209] predicted district energy demand from different levels, using machine-learning models. The machine-learning based model can provide prediction accuracy and adequate forecasting intervals in response to the requirement of smart grids. Wei et al. [33] systematically and comprehensively reviewed different data-driven based methods to predict building energy demand. The data-driven approaches are promising for the building energy study, including predictions on building demand and profile, district energy-consumption mapping, the benchmark of building clusters, retrofit strategies and guidelines. In respect to the robust model development, Sharifzadeh et al. [34] comparatively studied the robustness of different machine-learning methods, in terms of renewable energy prediction. The research results indicate that, compared to other models, the neural networks were more reliable for the prediction of electricity demand and renewable generations. AL-Musaylh et al. [35] investigated electric demand forecasting with an artificial neural network model. Research results indicate that, the hybrid ANN model generated a relative root-mean square error of around 3.85 %. In respect to a district energy network, Reynolds et al. [37] optimized the energy generation and demand management, by driving the artificial neural networks. Research results indicate that, through the optimisation, heat generation can be increased by 44.88 %, together with an increased profit of 8.04 %. Gallagher et al. [36] indicated that, the data-driven model can improve the accuracy for the uncertainty of energy-saving, with the reduced error at 51.09 %.

Fig. 10 demonstrates the evolution of papers using machine learning methods in energy-related research over the past ten years. The number of studies for machine learning applications in district energy systems has increased rapidly. As shown in Fig. 10(a) and (b), machine learning application for design and operation accounts for the largest share, whereas the proportion decreased from 52.1 % in 2010 to 40.4 % in 2019. The publication number on machine learning applications in performance prediction has increased from 106 to 1517, and the proportion from 6.0 % to 14.9 % over the last ten years. In addition, the machine-learning methods for the control strategy, energy management, uncertainty and sensitivity analysis have also increasingly attracted researchers' interest during the past ten years.

Furthermore, machine learning methods show promising prospects when integrated with optimisation algorithms for optimisation on nonlinear systems. The machine learning method can also be flexible to be integrated with newly developed optimisation algorithms. Byflexibly

Table 6
A holistic overview of optimal design and robust operation of district energy systems.

	Types	Study	Research/ Optimisation problem	Energy system scale	Solution/Algorithm	Objective(s)	DHC type
Parametric analysis	Renewable&distributed integrations	Manente et al. [26]	Single-objective	District heating system	Thermal storage in a district heating network	Electric power saving potential	Geothermal and waste-to-energy district heating system
	System design and planning	Rosato et al. [27]	Multi-objective	A small-scale Italian district (6 typical Italian residential buildings and 3 schools)	Robust power plant configurations based on decentralized heating system	Energy, emission and cost	A centralized solar hybrid heating system with seasonal storage
	System operation and control	Vanhoudt et al. [30]	Single-objective	District heating networks with different energy storage configurations	Cloud-based demand-response strategies	Investment costs	District heating networks and the effect of different thermal energy storage
System optimisation	Renewable&distributed integrations	Kazagic et al. [204]	Multi-objective	District	Modular district heating	Economic and environmental performances	District heating system with CHP and renewable energy sources
		Talebi et al. [28]	Multi-objective	Existing and newly built communities [28]	Dynamic optimisation	Annual life cycle cost and CO_2 emission	Hybrid community-district heating system
		Wang et al. [31]; Weinand et al. [32]	Single-objective	District	Genetic Algorithms [31] combinatorial optimisation approach (mixed-integer linear optimisation model and a three-stage heuristic) [32]	Annual net profit [31], minimum-cost [32]	Extraction-condensing CHP unit [31]; district heating networks [32]
	System design and planning	Tańczuk et al. [25]; Luo et al. [205]	Single-objective	Municipal district heating source [25]; district energy networks [205]	Techno-economical optimisation algorithm [25]; mixed integer linear programming [205]	Efficiency [25] and cost [205]	Gas turbine powered municipal district heating [25]; renewable supported district energy networks on a virtual island [205]
	System operation and control	Fazlollahi et al. [206]	Multi-objective	District energy systems	Mixed-integer linear programming	Environmental impact; total annual costs; penetration ratio of renewable energy [206]	Centralized and decentralized technologies
		Dorotić et al. [29]	Multi-objective	District	Linear programming	Discounted investment and operational costs; carbon dioxide emissions	District heating and cooling systems

integrating the machine learnt optimisation matrix in the teaching-learning-based optimisation algorithm, Zheng et al. [210] successfully identified the multivariable optimisation of an aerogel glazing system. According to the results, the total heat gain of an optimal system from the teaching-learning-based optimisation is 7.2 % lower than the total heat gain of an optimal solution through the particle swarm optimisation algorithm. The model predictive control can improve demand flexibility with great potentials in electricity cost saving and peak demand reduction.

4.3. Discussion on future prospects

Throughout the holistic and comprehensive literature review, the main concerns of machine learning techniques are on the optimal design and robust operation, control strategy and energy management, energy prediction, uncertainty and sensitivity analysis. Due to the intrinsic characteristics of computational efficiency with accurate feature extraction and classification, the upcoming avenues for future studies related to machine learning in district energy systems can be on.

1) surrogate model development to replace mathematical models without sacrificing the prediction accuracy;
2) short-term energy predictions for real-time energy control, and long-term prediction for model-predictive control;
3) generation of penalty signals from hybrid grids to improve techno-economic performances;
4) the fault detection and diagnosis of multi-energy systems in the entire energy chain, i.e., energy supply, transmission, transmission and distribution processes.

Several critical research challenges have been identified as avenues for future research. The identified challenges include the cost of a large amount of training data, the accuracy and robustness of developed machine learning model within advanced learning rules, synergic implementation of penalty signals from hybrid grids for the resilience and flexibility enhancement of districting energy network, hybrid grids' requirement in demand responsive and conflictions between different energy grids.

5. Outlook and recommendations

5.1. Technology implementations and practical applications

In the academia, researchers worldwide start to study machine learning-based digital district heating/cooling with renewable integrations and advanced low-carbon transition with preliminary research outcomes. For instance, Manfren et al. [211] studied data-driven building energy modelling with interpretability and generalisation. The approach can promote the scalability of data-driven techniques in district heating/cooling networks. For critical variables selection, Hwang et al. [212] proposed a systematic parameter selection approach based on domain knowledge and correlation analysis, promoting the prediction of electric energy consumption of heating and cooling systems. The cyber-physical systems and artificial intelligence can further promote the systemic resilience and energy security [213]. Meanwhile, research results in this study can also be widely applied in different climates. The cooling/heating network design, the demand and grid-responsive control, advanced energy conversions, and the flexible management can improve the energy resilience and flexibility, when

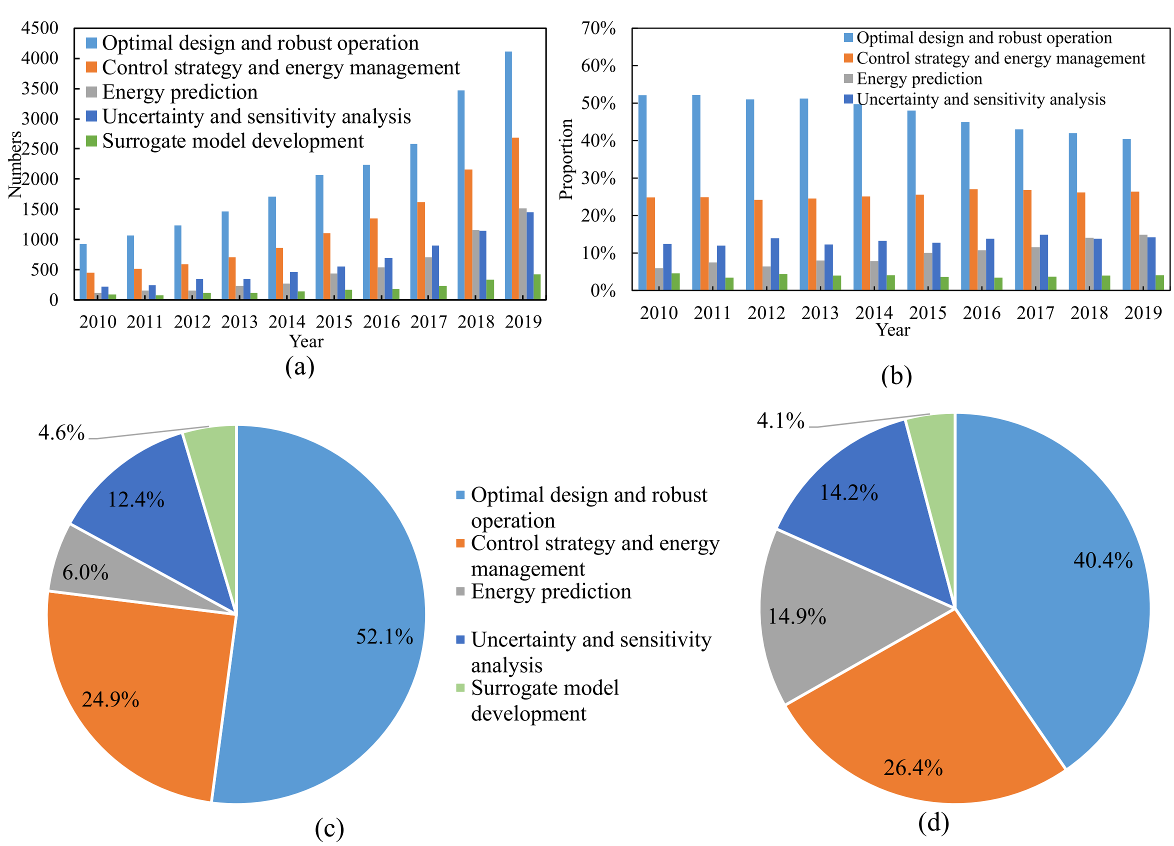


Fig. 10. Machine learning methods in district energy systems for different research purposes: (a) the number of research studies; (b) the proportion of research papers; (c) the constitution of studies in 2010; (d) the constitution of studies in 2019.

suffering from climate change or extreme events. Parametric analysis, single-objective and multi-objective optimizations can assist the optimal design and robust operation of district energy systems. Furthermore, machine learning technique can be used in the prediction of district energy demand and the robust model development, the surrogate model development for uncertainty and optimisation analyses, the geometrical and operating parameters' optimal design, and the efficient and robust energy management and control strategy.

The integration of various energy sources, storages (like plug-in EVs), heating/cooling/power/hydrogen grids makes the multi-energy system becomes complex in both structural configuration and accurate performance prediction. Challenging issues mainly include flexible interfaces for various components integration and temporal mismatch due to the simulation time inconsistency in convergence for each integrated component. For example, it will take around three days to predict the one day thermodynamic and energy output performance of an aerogel-based PV/T-PCM system [214] (due to the iteration of sunlight and heat transmission within the nanoporous aerogel structure, heat transfer of latent phase change materials via finite volume method and enthalpy-porosity), while the end-users (like building owners) do not allow so long time for calculation as the end-users only care about the real-time supplied power. The machine learning can address this issue through the multiple linear regression, the support vector regression and the backpropagation neural network. The overfitting is solved by eliminating the unnecessary connections with small weights regularization, adding the sum of the weights to the learning function. Generally, the machine learning based techniques can play significant roles in future integrated multi-energy systems due to high computational efficiency,

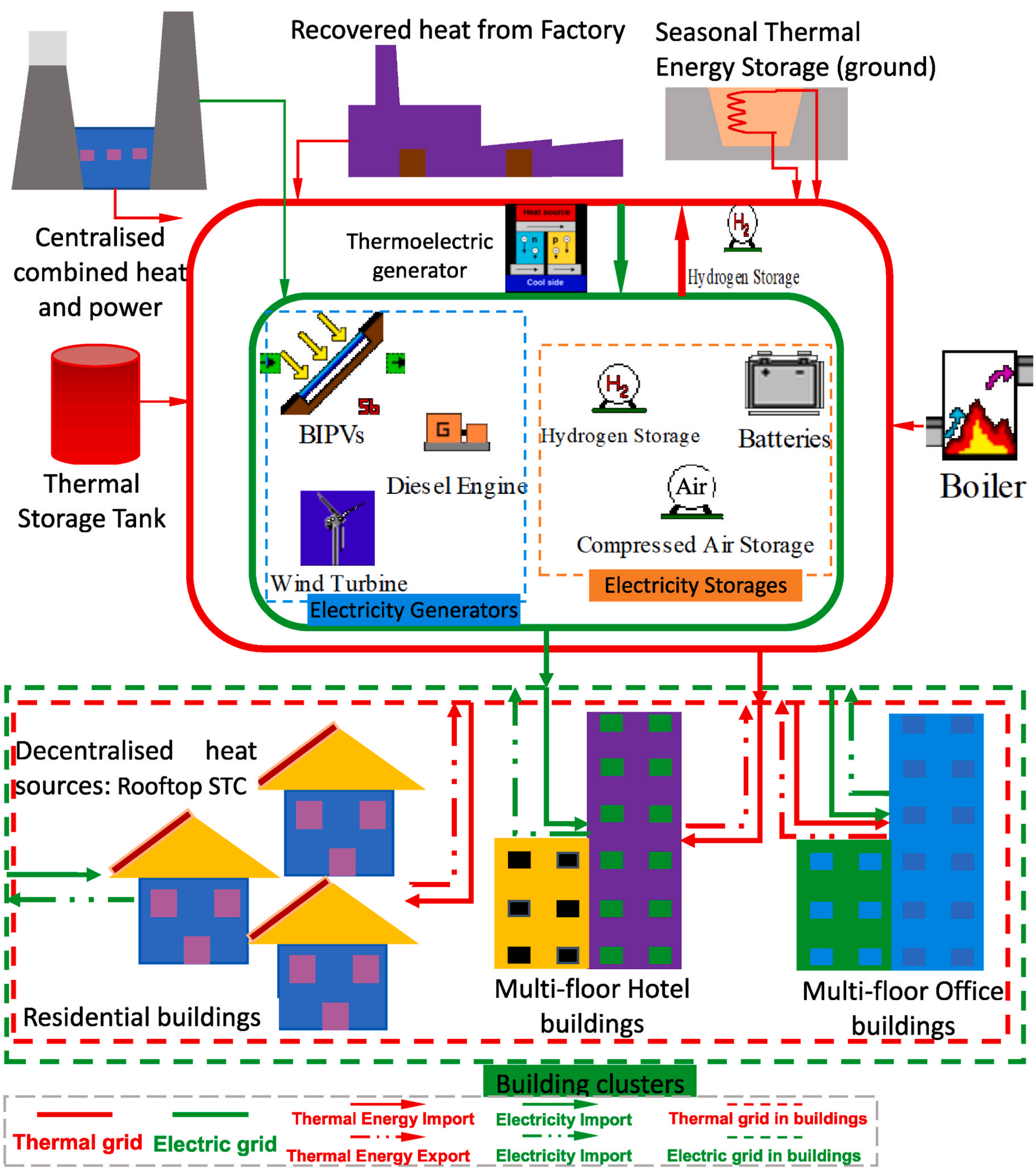


Fig. 11. A district energy system with renewable systems, diversified energy forms and hybrid energy storages..
(Note: the STC and BIPVs refer to solar thermal collector and building integrated photovoltaics.)

accurate performance prediction of nonlinear systems, user-friendly interface, including accurate performance prediction [215,216], advanced model predictive controls [217], single- and multi-objective optimisations [218,219].

5.2. Research limitations

Several research limitations of this study are summarized below.

1) advanced controls or strategies are seldomly applied in real systems, due to the user-unfriendly, heavy computational resources, mismatch between the model iteration and real-time reaction requirement of the system. For example, in the case when the computational time-duration for iteration until the convergence of model predictive controls is longer than the permitted system response time, the MPC cannot be effectively applied in cooling/heating systems;
2) enhancement strategies for energy resilience and flexibility of district heating and cooling networks have not considered the distributed energy system typology, smart interactions ad effectiveness assessment;
3) nexus among energy digitalisation, energy network typology, infrastructures and big data have not been clearly provided.

Furthermore, along with the planning, design and operation processes of integrated multi-energy systems, error sources are multi-diversified, like uncertainties from renewables, stochastic energy-use behaviours and vehicle driving behaviours, electricity prices, energy policies, and so on. These factors will affect the predicted technical performances, optimal results, and economic-environmental performances. Policy makers may need to initiate incentives to promote the infrastructure installation, widespread adoption of advanced science technologies. Furthermore, the advancement in machine learning can improve the performance prediction accuracy in high efficiency, through multiple linear regression, support vector regression and backpropagation neural network.

5.3. Challenges and future studies

Throughout the systematic and comprehensive literature review, district energy systems with interactive energy sharing mechanisms show promising prospects for renewable integration with cleaner power production, mitigating grids' dependence and global warming, and the resilience improvement of smart energy systems. Both district heating and cooling networks have been formulated, integrating energy storages and conversions for the cascade energy utilisation. Renewable sources include solar photovoltaics, solar thermal energy, geothermal energy, biomass and recovered waste energy. The role of coal, diesel and natural gas can thereafter be transited from main sources to the backup. Within the district energy system, thermal energy storages include sensible, latent and thermochemical energy storages. Electric storages include battery storage, compressed air energy storage, supercapacity, fly-wheel and hydrogen energy storage. Fig. 11 demonstrates a general working principle of district energy systems with renewable systems, diversified energy forms and hybrid energy storages. Centralised (combined heat and power) and decentralized energy sources (solar thermal collectors in buildings, building integrated photovoltaics and diesel engines) are included to support the district energy system. Waste heat recovery from the centralized factor accounts for a large share of all energy sources, promoting the renewable and sustainable development of smart energy systems. Multi-diversified energy storage systems include hydrogen, compressed air storage and battery for real-time and daily electricity storage, ground for seasonal thermal storage purposes. Energy conversion units, such as a thermoelectric generator for thermal-to-electric conversion and a heat pump for electric-to-thermal conversion, are installed for cascade energy sharing between different energy grids. From the perspective of the end-user, different types of buildings can be participated in the district energy system, including residential buildings, multi-floor hotel buildings and multi-floor office buildings.

In respect to the formulated district energy system, several challenges were presented for future studies.

1) Benchmarks to select the most suitable energy storage systems and advanced energy management strategies. Considering the mechanism difference of diversified energy storages and local conditions (such as climate and geographical conditions), appropriate energy storage systems can guide designers and stakeholders.
2) The conflictions of hybrid grids need to be well addressed, considering the local climates, users' needs and occupants' requirements. The synergic functions between different grids with diversified energy conversions can enhance the resilience and flexibility of hybrid grids for integrating intermittent renewable sources and covering stochastic demands.
3) The formulated district energy system involves multi-dimensional variables (such as the size of energy source and storage systems, the occupants' behavior, operating schedule of devices and so on), and multi-objectives (such as the energy efficiency, the initial and operational costs, the equivalent CO_2 emission, life cycle cost and emission, and so on). It is necessary to conduct optimisations to promote the development of renewable and sustainable district energy networks with high penetration of renewable energy.
4) The further update of advanced optimisation algorithms is critical to enhancing the overall performance of district energy systems.
5) The multi-criteria decision-making is quite complicated for all stakeholders participating in the district energy systems, including the end-users, building owners, facility managers, aggregators, district and transmission operators. Developing effective tools to help stakeholders make the 'best of the best' decision, along the Pareto optimal front is necessary, to improve their participation willingness and readiness.
6) Environmental, Science and Government (ESG) policies to cultivate associated technologies and promote its widespread applications in low-carbon district heating/cooling.

6. Conclusions

In this study, district energy systems have been systematically and comprehensively presented, in respect to district heating/cooling networks, hybrid renewables' integration, energy storages (such as thermal and electric storages) with heating, cooling and electrical energy forms and advanced energy conversions (such as the solar-to-heat, the solar-to-electricity, the heating-to-cooling conversions), and hybrid grids' interactions (such as district cooling/heating grids, power grid, natural gas grid, biofuel grid and hydrogen grid). An in-depth dialectical analysis on advantages and disadvantages, together with the techno-economic feasibility, has been provided, in respect to diversified thermal and electrical storages. Energy management and control strategy have been reviewed from perspectives of the design on district energy networks, the demand and grid-responsive control, and the flexible management of hybrid energy storages, in accordance with hybrid grids' requirements. Roles of hybrid renewables, novel energy storage systems, and diversified grids' interactions have been analysed. The main function of diversified grids' interactions is to realise the dynamic balance between hybrid renewables and energy demands and improve the grids' resilience to fluctuations and intermittence of renewable energy supply. Both single-objective and multi-objective optimizations of district energy systems have been comprehensively presented to promote smart district energy systems with optimal design, robust operation, energy controls and resilient grids' interactions. Furthermore, due to intrinsic characteristics of computational efficiency with accurate feature extraction and classification, machine learning methods show promising prospects for the realisation towards resilient and smart district energy systems, in terms of prediction of district energy demands and robust model development, surrogate model development for uncertainty and optimisation analyses, optimal design on geometrical and operating parameters, efficient and robust energy management and control strategy. Due to the computational complexity of multivariable and multi-objective optimisations of district systems, machine learning methods can also be used to develop surrogate models to significantly reduce the computational load without sacrificing the accuracy of optimisation results. Last but not the least, a resilient and smart district energy network has been proposed to guide the design in the future city, involving on-site renewable generations, waste heat recovery from centralised power plants, multi-diversified energy storages, advanced energy conversions for the interactive energy sharing. Critical research results are summarized below.

1) In academia, energy storages in district energy systems include thermal energy storages (sensible, latent and thermochemical storages), and electricity storages (electrochemical battery, compressed air energy storage, supercapacitor, fly-wheel and hydrogen). Compared to latent or sensible heat storages, superiorities of thermochemical storages include higher energy density, higher operating temperature and higher efficiency. However, due to the high operating temperature, technical challenges include sophisticated system control, service life, material compatibility issues and chemical reactions. In respect to electricity storage, technical challenges include performance depreciation over long-time operations, operational security, enhancement of energy storage density, and energy loss during charging process. Furthermore, energy consumption from transportations needs to be considered when formulating smart district energy systems, in which the regenerative brake system is one of reliable and promising energy sources when integrating thermal and battery storages. The mobility of vehicles can improve regional renewable energy balance by overcoming uneven spatiotemporal distribution through smart charging/discharging technologies and energy sharing.
2) In order to promote resilient and smart district energy systems, district heating and cooling networks with diversified grids' interactions have been formulated. Technical challenges include the diversity of occupants' behaviours within different types of buildings, the consideration of hybrid grids' requirements and conflictions between different grids, and the peak power mitigation when enlarging the energy boundary from single building to a large scale, such as building cluster levels.
3) In order to improve the resilience and flexibility of district heating and cooling networks, technical solutions can be focused on the heating networks' design, the demand and grid-responsive control, and the flexible management of hybrid energy storages in accordance with hybrid grids' requirements. Furthermore, the activations of synergistic functions, considering differences of hybrid grids' requirement with diversified energy conversions, are effective solutions for enhancing the resilience and flexibility of multi-energy systems. From the perspective of heating networks' design, the district heating network with annular pipes is more reliable and flexible than the branched pipe network, in terms of the power supply and the mitigation on grids' pressure.
4) The main function of diversified grids' interactions is to dynamically balance the hybrid renewables and energy demands. Furthermore, the grids' resilience to fluctuations in the energy supply can be improved through advanced energy conversions between different energy grids in district energy systems. In respect to energy management, both demand-side management and energy storages can be operated to manage the renewable energy, in according to climate conditions, users' needs and grids' requirements. The availability of advanced energy conversions can promote the renewable penetration, energy sharing and grid' independence, in respect to diversified energy forms.
5) Parametric analysis, single-objective and multi-objective optimizations of district energy systems for the optimal design and robust operation have been investigated in the academia. The assessment criteria include system efficiency, electric power saving, primary energy consumption, equivalent CO_2 emissions, economic considerations (such as investment cost and operational cost) and life-cycle analysis. The adopted optimisation algorithms include the linear programming, the genetic algorithms, and the hybrid mixed-integer linear with a three-stage heuristic. However, due to the computational complexity and time-consuming of traditional optimisation methodology with the searching principle for all samples, the development of new optimisation engines to reduce the computational load without sacrificing the accuracy has attracted researchers' interest.
6) The interdisciplinary between the machine learning and district energy systems shows promising prospects for the promotions of resilient and smart district energy systems. The machine learning technique can be used in the prediction of district energy demand and the robust model development, the surrogate model development for uncertainty and optimisation analyses, the geometrical and operating parameters' optimal design, and the efficient and robust energy management and control strategy. Furthermore, in respect to the computational complexity and the time-consuming of optimisations, trade-off strategies between computational complexity and accuracy can be reached, through the operation of machine learning approach with optimisation algorithms.
7) In respect to the formulated district energy system with high energy flexibility and resilience, identified technical challenges include benchmarks for the selection of most suitable energy storage systems considering the mechanism difference of diversified energy storages and local conditions (such as climate and geographical conditions), conflictions of hybrid grids, optimisations with advanced optimisation algorithms, and multi-criteria decision-making to promote the willingness and readiness for stakeholders' participations.

The literature review and rational analysis show that renewables' integration, advanced thermal and electric energy storages, energy management and control strategy, and diversified grids' interactions, are critical elements of the resilient and smart district energy systems. Machine learning methods are promising. Future work can focus on the enhancement of resilience and flexibility of multi-energy systems with activations of synergistic functions of hybrid grids with diversified energy conversions, the generation of penalty signals from hybrid grids for the improvement of techno-economic performances, the flexible transition from physically and commercially isolated district energy systems to interactive energy sharing networks, and the machine-learning based optimisation methodology flexibly integrating advanced optimisation algorithms. This review can contribute to the regulation/standards to policy makers, applications, and policy targets in industry, countries/regions, considering energy, emissions/environment, costs, climate change, Environmental, Science and Government (ESG) and Sustainable Development Goals (SDGs).

CRediT authorship contribution statement

Yuekuan Zhou: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Siqian Zheng:** Validation, Visualization, Writing – original draft, Writing – review & editing. **Jan L.M. Hensen:** Validation, Visualization, Revison.

Declaration of competing interest

The authors declared that they have no conflicts of interest to this work. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Data availability

The data that has been used is confidential.

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References

[1] Zhang S, Chen W. China's energy transition pathway in a carbon neutral vision. Engineering 2022;14:64–76.
[2] Zhou Y, Zheng S, Lei J. A cross-scale modelling and decarbonisation quantification approach for navigating Carbon Neutrality Pathways in China. Energy Convers Manag 2023;297:117733.
[3] Zhou Y. Low-carbon transition in smart city with sustainable airport energy ecosystems and hydrogen-based renewable-grid-storage-flexibility. Energy Rev 2022;1(1):100001.
[4] Zhou Y, Zheng S. A co-simulated material-component-system-district framework for climate-adaption and sustainability transition. Renew Sustain Energy Rev 2024;192:114184.
[5] Zhou Y, Liu Z. A cross-scale 'material-component-system' framework for transition towards zero-carbon buildings and districts with low, medium and high-temperature phase change materials. Sustain Cities Soc 2023;89:104378.
[6] Zhou Y. Transition towards carbon-neutral districts based on storage techniques and spatiotemporal energy sharing with electrification and hydrogenation. Renew Sustain Energy Rev 2022;162:112444.
[7] Zhou Y. Climate change adaptation with energy resilience in energy districts—A state-of-the-art review. Energy Build 2023;279:112649.
[8] Zhou Y, Dan Z, Yu X. Climate-adaptive resilience in district buildings and cross-regional energy sharing in Guangzhou-Shenzhen-Hong Kong Greater Bay Area. Energy Build 2024;308:114004.
[9] Zhou Y, Liu X. A stochastic vehicle schedule model for demand response and grid flexibility in a renewable-building-e-transportation-microgrid. Renew Energy 2024;221:119738.
[10] Zhou Y. Sustainable energy sharing districts with electrochemical battery degradation in design, planning, operation and multi-objective optimisation. Renew Energy 2023;202:1324–41.
[11] Dan Z, Song A, Yu X. Electrification-driven circular economy with machine learning-based multi-scale and cross-scale modelling approach. Energy 2024;299:131469.
[12] Henry A, Prasher R, Majumdar A. Five thermal energy grand challenges for decarbonization. Nat Energy 2020;5:635–7.
[13] Thiel GP, Stark AK. To decarbonize industry, we must decarbonize heat. Joule 2021;5(3):531–50.
[14] Allegrini J, Orehounig K, Mavromatidis G, Rueschm F, Dorer V, Evins R. A review of modelling approaches and tools for the simulation of district-scale energy systems. Renew Sustain Energy Rev 2015;52:1391–404.
[15] Gross R, Hann R. Path dependency in provision of domestic heating. Nat Energy 2019;4:358–64.
[16] Barrington-Leigh C, Baumgartner J, Carter E, Robinson BE, Tao S, Zhang Y. An evaluation of air quality, home heating and well-being under Beijing's programme to eliminate household coal use. Nat Energy 2019;4:416–23.
[17] Kim J, Sengodan S, Kim S, Kwon O, Bu Y, Kim G. Proton conducting oxides: a review of materials and applications for renewable energy conversion and storage. Renew Sustain Energy Rev 2019;109:606–18.
[18] Lee I, Tester JW, You F. Systems analysis, design, and optimization of geothermal energy systems for power production and polygeneration: state-of-the-art and future challenges. Renew Sustain Energy Rev 2019;109:551–77.
[19] Carrillo AJ, González-Aguilar J, Romero M, Coronado JM. Solar energy on demand: a review on high temperature thermochemical heat storage systems and materials. Chem Rev 2019;119(7):4777–816.
[20] Alanne K, Cao S. Zero-energy hydrogen economy (ZEH2E) for buildings and communities including personal mobility. Renew Sustain Energy Rev 2017;71:697–711.
[21] Amy C, Seyf HR, Steiner MA, Friedman DJ, Henry A. Thermal energy grid storage using multi-junction photovoltaics. Energy Environ Sci 2019;12:334–43.
[22] Di Silvestre ML, Favuzza S, Sanseverino ER, Zizzo G. How Decarbonization, Digitalization and Decentralization are changing key power infrastructures. Renew Sustain Energy Rev 2018;93:483–98.
[23] Inayat A, Raza M. District cooling system via renewable energy sources: a review. Renew Sustain Energy Rev 2019;107:360–73.
[24] Zhou Y. Worldwide carbon neutrality transition? Energy efficiency, renewable, carbon trading and advanced energy policies. Energy Rev 2023;2(2):100026.
[25] Tańczuk M, Skorek J, Bargiel P. Energy and economic optimization of the repowering of coal-fired municipal district heating source by a gas turbine. Energy Convers Manag 2017;149:885–95.
[26] Manente G, Lazzaretto A, Molinari I, Bronzini F. Optimization of the hydraulic performance and integration of a heat storage in the geothermal and waste-to-energy district heating system of Ferrara. J Clean Prod 2019;230:869–87.
[27] Rosato A, Ciervo A, Ciampi G, Scorpio M, Sibilio S. Impact of seasonal thermal energy storage design on the dynamic performance of a solar heating system serving a small-scale Italian district composed of residential and school buildings. J Energy Storage 2019. https://doi.org/10.1016/j.est.2019.100889.
[28] Talebi B, Haghighat F, Tuohy P, Mirzaei PA. Optimization of a hybrid community district heating system integrated with thermal energy storage system. J Energy Storage 2019;23:128–37.
[29] Dorotić H, Pukšec T, Duić N. Multi-objective optimization of district heating and cooling systems for a one-year time horizon. Energy 2019;169:319–28.
[30] Vanhoudt D, Claessens BJ, Salenbien R, Desmedt J. An active control strategy for district heating networks and the effect of different thermal energy storage configurations. Energy Build 2018;158:1317–27.
[31] Wang J, Zhou Z, Zhao J, Zheng J, Guan Z. Optimizing for clean-heating improvements in a district energy system with high penetration of wind power. Energy 2019;175:1085–99.
[32] Weinand JM, Kleinebrahm M, McKenna R, Mainzer K, Fichtner W. Developing a combinatorial optimisation approach to design district heating networks based on deep geothermal energy. Appl Energy 2019. https://doi.org/10.1016/j.apenergy.2019.113367.
[33] Wei Y, Zhang X, Shi Y, Xia L, Pan S, Wu J, Han M, Zhao X. A review of data-driven approaches for prediction and classification of building energy consumption. Renew Sustain Energy Rev 2017;82:1027–47.
[34] Sharifzadeh M, Sikinioti-Lock A, Shah N. Machine-learning methods for integrated renewable power generation: a comparative study of artificial neural networks, support vector regression, and Gaussian Process Regression. Renew Sustain Energy Rev 2019;108:513–38.
[35] AL-Musaylh MS, Deo RC, Adamowski JF, Li Y. Short-term electricity demand forecasting using machine learning methods enriched with ground-based climate and ECMWF Reanalysis atmospheric predictors in southeast Queensland, Australia. Renew Sustain Energy Rev 2019. https://doi.org/10.1016/j.rser.2019.109293.
[36] Gallagher CV, Bruton K, Leahy K, Sulliv DTJO. The suitability of machine learning to minimise uncertainty in the measurement and verification of energy savings. Energy Build 2018;158:647–55.
[37] Reynolds J, Ahmad MW, Rezgui Y, Hippolyte JL. Operational supply and demand optimisation of a multi-vector district energy system using artificial neural networks and a genetic algorithm. Appl Energy 2019;235:699–713.
[38] Zhou Y. A dynamic self-learning grid-responsive strategy for battery sharing economy—multi-objective optimisation and posteriori multi-criteria decision making. Energy 2023;266:126397.
[39] Chu S, Majumdar A. Opportunities and challenges for a sustainable energy future. Nature 2012;488:294–303.
[40] Gur I, Sawyer K, Prasher R. Searching for a better thermal battery. Science 2012;335(6075):1454–5.
[41] Zhou Y. A dynamic self-learning grid-responsive strategy for battery sharing economy—multi-objective optimisation and posteriori multi-criteria decision making. Energy 2023;266:126397.
[42] Amirioun MH, Kazemi A. A new model based on optimal scheduling of combined energy exchange modes for aggregation of electric vehicles in a residential complex. Energy 2014;69(5):186–98.
[43] Zhang Y, Wang R, Zhang T, Liu Y, Guo B. Model predictive control-based operation management for a residential microgrid with considering forecast uncertainties and demand response strategies. IET Gener Transm Distrib 2016;10(10):2367–78.

[44] Liu Z, Sun P, Li S, Yu Z, Mankibi ME, Roccamena L, Yang T, Zhang G. Enhancing a vertical earth-to-air heat exchanger system using tubular phase change material. J Clean Prod 2019. https://doi.org/10.1016/j.jclepro.2019.117763.
[45] Islam MS. A techno-economic feasibility analysis of hybrid renewable energy supply options for a grid-connected large office building in southeastern section of France. Sustain Cities Soc 2018;38:492–508.
[46] Li X, Ogden J, Yang C. Analysis of the design and economics of molten carbonate fuel cell tri-generation systems providing heat and power for commercial buildings and H2 for FC vehicles. J Power Sources 2013;241(11):668–79.
[47] Knobloch F, Hanssen S, Lam A, et al. Net emission reductions from electric cars and heat pumps in 59 world regions over time. Nat Sustain 2020;3:437–47.
[48] Flores RJ, Shaffer BP, Brouwer J. Electricity costs for a Level 3 electric vehicle fueling station integrated with a building. Appl Energy 2017;191:367–84.
[49] Ioakimidis CS, Thomas D, Rycerski P, Genikomsakis KN. Peak shaving and valley filling of power consumption profile in non-residential buildings using an electric vehicle parking lot. Energy 2018;148:148–58.
[50] Sehar F, Pipattanasomporn M, Rahman S. Demand management to mitigate impacts of plug-in electric vehicle fast charge in buildings with renewables. Energy 2016;120:642–51.
[51] Yang Z, Garimella SV. Thermal analysis of solar thermal energy storage in a molten-salt thermocline. Sol Energy 2010;84(6):974–85.
[52] Mazhar AR, Liu S, Shukla A. A state of art review on the district heating systems. Renew Sustain Energy Rev 2018;96:420–39.
[53] Wang G, Wang W, Luo J, Zhang Y. Assessment of three types of shallow geothermal resources and ground-source heat-pump applications in provincial capitals in the Yangtze River Basin, China. Renew Sustain Energy Rev 2019;111: 392–421.
[54] Herrmann U, Kelly B, Price H. Two-tank molten salt storage for parabolic trough solar power plants. Energy 2004;29(5–6):883–93.
[55] Yang Z, Garimella SV. Cyclic operation of molten-salt thermal energy storage in thermoclines for solar power plants. Appl Energy 2013;103:256–65.
[56] Zhou Y, Zheng S, Zhang G. Artificial neural network based multivariable optimization of a hybrid system integrated with phase change materials, active cooling and hybrid ventilations. Energy Convers Manag 2019. https://doi.org/10.1016/j.enconman.2019.111859.
[57] Zhou Y, Zheng S, Zhang G. Study on the energy performance enhancement of a new PCMs integrated hybrid system with the active cooling and hybrid ventilations. Energy 2019;179:111–28.
[58] Scapino L, Zondag HA, Van Bael J, Diriken J, Rindt CCM. Sorption heat storage for long-term low-temperature applications: a review on the advancements at material and prototype scale. Appl Energy 2017;190:920–48.
[59] Scapino L, De Servi C, Zondag HA, Diriken J, Rindt CCM, Sciacovelli A. Techno-economic optimization of an energy system with sorption thermal energy storage in different energy markets. Appl Energy 2020;258:114063.
[60] Milis K, Peremans H, Springael J, Passel SV. Win-win possibilities through capacity tariffs and battery storage in microgrids. Renew Sustain Energy Rev 2019. https://doi.org/10.1016/j.rser.2019.06.045.
[61] Wang F, Zhan X, Cheng Z, Wang Z, Wang Q, Xu K, Safdar M, He J. Tungsten oxide @ polypyrrole core–shell nanowire arrays as novel negative electrodes for asymmetric supercapacitors. Small 2014. https://doi.org/10.1002/smll.201402340.
[62] Muzaffar A, Ahamed MB, Deshmukh K, Thirumalai J. A review on recent advances in hybrid supercapacitors: design, fabrication and applications. Renew Sustain Energy Rev 2019;101:123–45.
[63] Zhang L, Hu X, Wang Z, Sun F, Dorrell DG. A review of supercapacitor modeling, estimation, and applications: a control/management perspective. Renew Sustain Energy Rev 2018;81:1868–78.
[64] Mousavi GSM, Faraji F, Majazi A, Al-Haddad K. A comprehensive review of flywheel energy storage system technology. Renew Sustain Energy Rev 2017;67: 477–90.
[65] He W, Wang J. Optimal selection of air expansion machine in Compressed Air Energy Storage: a review. Renew Sustain Energy Rev 2018;87:77–95.
[66] Venkataramani G, Parankusam P, Ramalingam V, Wang J. A review on compressed air energy storage – a pathway for smart grid and polygeneration. Renew Sustain Energy Rev 2016;62:895–907.
[67] Marvania D, Subudhi S. A comprehensive review on compressed air powered engine. Renew Sustain Energy Rev 2017;70:1119–30.
[68] Karellas S, Tzouganatos N. Comparison of the performance of compressed-air and hydrogen energy storage systems: karpathos island case study. Renew Sustain Energy Rev 2014;29:865–82.
[69] Parra D, Valverde L, Pino FJ, Patel MK. A review on the role, cost and value of hydrogen energy systems for deep decarbonisation. Renew Sustain Energy Rev 2019;101:279–94.
[70] Balcombe P, Speirs J, Johnson E, Martin J, Brandon N, Hawkes A. The carbon credentials of hydrogen gas networks and supply chains. Renew Sustain Energy Rev 2018;91:1077–88.
[71] González-Roubaud E, Pérez-Osorio D, Prieto C. Review of commercial thermal energy storage in concentrated solar power plants: steam vs. molten salts. Renew Sustain Energy Rev 2017;80:133–48.
[72] Kazimierczuk AH. Wind energy in Kenya: a status and policy framework review. Renew Sustain Energy Rev 2019;107:434–45.
[73] Liu ZX, Yu Z, Yang TT, Zhang GQ, Haghighat F, Joybari MM. A review on macro-encapsulated phase change material for building envelope applications. Build Environ 2018;144:281–94.
[74] Buffat R, Raubal M. Spatio-temporal potential of a biogenic micro CHP swarm in Switzerland. Renew Sustain Energy Rev 2019;103:443–54.
[75] Präger F, Paczkowski S, Sailer G, Derkyi NSA, Pelz S. Biomass sources for a sustainable energy supply in Ghana – a case study for Sunyani. Renew Sustain Energy Rev 2019;107:413–24.
[76] Hoehne CG, Chester MV. Optimizing plug-in electric vehicle and vehicle-to-grid charge scheduling to minimize carbon emissions. Energy 2016;115:646–57.
[77] Nunes P, Brito MC. Displacing natural gas with electric vehicles for grid stabilization. Energy 2017;141:87–96.
[78] Gravelsins A, Pakere I, Tukulis A, Blumberga D. Solar power in district heating. P2H flexibility concept. Energy 2019;181:1023–35.
[79] Wang S, Tang R. Supply-based feedback control strategy of air-conditioning systems for direct load control of buildings responding to urgent requests of smart grids. Appl Energy 2017;201:419–32.
[80] Lawrence TM, Boudreau MC, Helsen L, Henze G, Mohammadpour J, Noonan D, Patteeuw D, Pless S, Watson RT. Ten questions concerning integrating smart buildings into the smart grid. Build Environ 2016;108(11):273–83.
[81] Schill WP, Zerrahn A. Long-run power storage requirements for high shares of renewables: results and sensitivities. Renew Sustain Energy Rev 2017;83:156–71.
[82] Khalaj AH, Abdulla K, Halgamuge SK. Towards the stand-alone operation of data centers with free cooling and optimally sized hybrid renewable power generation and energy storage. Renew Sustain Energy Rev 2018;93:451–72.
[83] Fernández RÁ. A more realistic approach to electric vehicle contribution to greenhouse gas emissions in the city. J Clean Prod 2018;172:949–59.
[84] Turkmen AC, Solmaz S, Celik C. Analysis of fuel cell vehicles with advisor software. Renew Sustain Energy Rev 2017;70:1066–71.
[85] Thomas D, Deblecker O, Ioakimidis CS. Optimal operation of an energy management system for a grid-connected smart building considering photovoltaics' uncertainty and stochastic electric vehicles' driving schedule. Appl Energy 2018;210:1188–206.
[86] Colmenar-Santos A, Linares-Mena AR, Borge-Diez D, Quinto-Alemany CD. Impact assessment of electric vehicles on islands grids: a case study for Tenerife (Spain). Energy 2017;120:385–96.
[87] Bulut MB, Odlare M, Stigson P, Wallin F, Vassileva I. Active buildings in smart grids—exploring the views of the Swedish energy and buildings sectors. Energy Build 2016;117:185–98.
[88] Rahbari O, Vafaeipour M, Omar N, Rosen MA, Hegazy O, Timmermans JM, Heibati S, Bossche PVD. An optimal versatile control approach for plug-in electric vehicles to integrate renewable energy sources and smart grids. Energy 2017;134: 1053–67.
[89] Alirezaei M, Noori M, Tatari O. Getting to net zero energy building: investigating the role of vehicle to home technology. Energy Build 2016;130:465–76.
[90] Wang Z, Wang L, Dounis AI, Yang R. Integration of plug-in hybrid electric vehicles into energy and comfort management for smart building. Energy Build 2012;47(47):260–6.
[91] Bolwig S, Bazbauers G, Klitkou A, Lund PD, Blumberga A, Gravelsins A, Blumberga D. Review of modelling energy transitions pathways with application to energy system flexibility. Renew Sustain Energy Rev 2019;101:440–52.
[92] Lu Y, Tian Z, Peng P, Niu J, Li W, Zhang H. GMM clustering for heating load patterns in-depth identification and prediction model accuracy improvement of district heating system. Energy Build 2019;190:49–60.
[93] Aslan A, Yüksel B, Akyol T. Effects of different operating conditions of Gonen geothermal district heating system on its annual performance. Energy Convers Manag 2014;79:554–67.
[94] Ratismith W, Favre Y, Canaff M, Briggs J. A non-tracking concentrating collector for solar thermal applications. Appl Energy 2017;200:39–46.
[95] Qu W, Wang R, Hong H, Sun J, Jin H. Test of a solar parabolic trough collector with rotatable axis tracking. Appl Energy 2017;207:7–17.
[96] Ruilin W, Wanjun Q, Hui H, Sun J, Jin H. Experimental performance of 300 kW th, prototype of parabolic trough collector with rotatable axis and irreversibility analysis. Energy 2018;161:595–609.
[97] Sorknæs P. Simulation method for a pit seasonal thermal energy storage system with a heat pump in a district heating system. Energy 2018;152:533–8.
[98] Liu M, Wang S, Zhao Y, Tang H, Yan J. Heat–power decoupling technologies for coal-fired CHP plants: operation flexibility and thermodynamic performance. Energy 2019. https://doi.org/10.1016/j.energy.2019.116074.
[99] Zhao S, Ge Z, Sun J, Ding Y, Yang Y. Comparative study of flexibility enhancement technologies for the coal-fired combined heat and power plant. Energy Convers Manag 2019;184:15–23.
[100] Li H, Wang B, Yan J, Salman CA, Thorin E, Schwede S. Performance of flue gas quench and its influence on biomass fueled CHP. Energy 2019;180:934–45.
[101] Akhtari S, Sowlati T, Day K. Economic feasibility of utilizing forest biomass in district energy systems – a review. Renew Sustain Energy Rev 2014;33:117–27.
[102] Singh K, Hachem-Vermette C. Influence of mixed-use neighborhood developments on the performance of waste-to-energy CHP plant. Energy 2019. https://doi.org/10.1016/j.energy.2019.116172.
[103] Davies G, Boot-Handford N, Curry D, Dennis W, Ajileye A, Revesz A, Maidment G. Combining cooling of underground railways with heat recovery and reuse. Sustain Cities Soc 2019;45:543–52.
[104] Revesz A, Chaer I, Thompson J, Mavroulidou M, Gunn M, Maidment G. Modelling of heat energy recovery potential from underground railways with nearby vertical ground heat exchangers in an urban environment. Appl Therm Eng 2019;147: 1059–69.
[105] Buonomano A, Calise F, Palombo A. Solar heating and cooling systems by absorption and adsorption chillers driven by stationary and concentrating photovoltaic/thermal solar collectors: modelling and simulation. Renew Sustain Energy Rev 2018;82:1874–908.

[106] Lizana J, Chacartegui R, Barrios-Padura A, Ortizc C. Advanced low-carbon energy measures based on thermal energy storage in buildings: a review. Renew Sustain Energy Rev 2018;82:3705–49.
[107] Kohlhepp P, Harb H, Wolisz H, Waczowicz S, Müller D, Hagenmeyer V. Large-scale grid integration of residential thermal energy storages as demand-side flexibility resource: a review of international field studies. Renew Sustain Energy Rev 2019;101:527–47.
[108] SDH. Solar-DH plants. http://solar-district-heating.eu/Documents/SDHCasestudies.asp; 2018.
[109] Xu X, You S, Zheng X, Li H. A survey of district heating systems in the heating regions of northern China. Energy 2014;77:909–25.
[110] Huang J, Fan J, Furbo S. Feasibility study on solar district heating in China. Renew Sustain Energy Rev 2019;108:53–64.
[111] Pospíšil J, Charvát P, Arsenyeva O, Klimeš L, Špiláček M, Klemeš JJ. Energy demand of liquefaction and regasification of natural gas and the potential of LNG for operative thermal energy storage. Renew Sustain Energy Rev 2019;99:1–15.
[112] Gong M, Werner S. Exergy analysis of network temperature levels in Swedish and Danish district heating systems. Renew Energy 2015;84:106–13.
[113] Wang J, Wang Z, Zhou D, Sun K. Key issues and novel optimization approaches of industrial waste heat recovery in district heating systems. Energy 2019. https://doi.org/10.1016/j.energy.2019.116005.
[114] Lu D, Chen G, Gong M, Bai Y, Xu Q, Zhao Y, Dong X, Shen J. Thermodynamic and economic analysis of a gas-fired absorption heat pump for district heating with cascade recovery of flue gas waste heat. Energy Convers Manag 2019;185:87–100.
[115] Rudra S, Tesfagaber YK. Future district heating plant integrated with municipal solid waste (MSW) gasification for hydrogen production. Energy 2019;180:881–92.
[116] Fabre A, Thomas R, Duplessis B, Tran CT, Stabat P. Dynamic modeling for evaluation of triple-pipe configuration potential in geothermal district heating networks. Energy Convers Manag 2018;173:461–9.
[117] Sun F, Zhao J, Fu L, Sun J, Zhang S. New district heating system based on natural gas-fired boilers with absorption heat exchangers. Energy 2017;138:405–18.
[118] Agbor E, Zhang X, Kumar A. A review of biomass co-firing in North America. Renew Sustain Energy Rev 2014;40:930–43.
[119] Buongiorno J, Carmichael B, Dunkin B, Parsons J, Smit D. Can nuclear batteries Be economically competitive in large markets? Energies 2021;14(14):4385.
[120] Jasserand F, de Lavergne JGD. Initial economic appraisal of nuclear district heating in France. EPJ Nuclear Sciences & Technologies; 2016.
[121] Khosravi A, Olkkonen V, Farsaei A, Syri S. Replacing hard coal with wind and nuclear power in Finland-impacts on electricity and district heating markets. Energy 2020;203:117884.
[122] Leurent M, Da Costa P, Jasserand F, Rämä M, Persson U. Cost and climate savings through nuclear district heating in a French urban area. Energy Pol 2018;115:616–30.
[123] Wahlroos M, Pärssinen M, Manner J, Syri S. Utilizing data center waste heat in district heating–Impacts on energy efficiency and prospects for low-temperature district heating networks. Energy 2017;140:1228–38.
[124] Ebrahimi K, Jones GF, Fleischer AS. A review of data center cooling technology, operating conditions and the corresponding low-grade waste heat recovery opportunities. Renew Sustain Energy Rev 2014;31:622–38.
[125] Yu J, Jiang Y, Yan Y. A simulation study on heat recovery of data center: a case study in Harbin, China. Renew Energy 2019;130:154–73.
[126] Lund H, Duic N, Østergaard PA, Mathiesen BV. Future district heating systems and technologies: on the role of smart energy systems and 4th generation district heating. Energy 2018;165:614–9.
[127] Li X, Song J, Yu G, Liang Y, Tian H, Shu G, Markides CN. Organic Rankine cycle systems for engine waste-heat recovery: heat exchanger design in space-constrained applications. Energy Convers Manag 2019. https://doi.org/10.1016/j.enconman.2019.111968.
[128] Shi L, Shu G, Tian H, Deng S. A review of modified Organic Rankine cycles (ORCs) for internal combustion engine waste heat recovery (ICE-WHR). Renew Sustain Energy Rev 2018;92:95–110.
[129] García NP, Zubi G, Pasaoglu G, Dufo-López R. Photovoltaic thermal hybrid solar collector and district heating configurations for a Central European multi-family house. Energy Convers Manag 2017;148:915–24.
[130] Tian Z, Perers B, Furbo S, Fan J. Thermo-economic optimization of a hybrid solar district heating plant with flat plate collectors and parabolic trough collectors in series. Energy Convers Manag 2018;165:92–101.
[131] Simpson MC, Chatzopoulou MA, Oyewunmi OA, Brun NL, Sapin P, Markides CN. Technoeconomic analysis of internal combustion engine–organic Rankine cycle systems for combined heat and power in energy-intensive buildings. Appl Energy 2019;253:113462.
[132] Block C, Ephraim A, Weiss-Hortala E, Minh DP, Nzihou A, Vandecasteele C. Co-pyrogasification of plastics and biomass, a review. Waste and Biomass Valorization 2019;10:483–509.
[133] Li G. Sensible heat thermal storage energy and exergy performance evaluations. Renew Sustain Energy Rev 2016;53:897–923.
[134] Bott C, Dressel I, Bayer P. State-of-technology review of water-based closed seasonal thermal energy storage systems. Renew Sustain Energy Rev 2019. https://doi.org/10.1016/j.rser.2019.06.048.
[135] Song A, Zhou Y. Advanced cycling ageing-driven circular economy with E-mobility-based energy sharing and lithium battery cascade utilisation in a district community. J Clean Prod 2023;415:137797.
[136] D'Ettorre F, Banaei M, Ebrahimy R, et al. Exploiting demand-side flexibility: state-of-the-art, open issues and social perspective. Renew Sustain Energy Rev 2022;165:112605.
[137] Gao B. In-field experimental study and multivariable analysis on a PEMFC combined heat and power cogeneration for climate-adaptive buildings with taguchi method. Energy Convers Manag 2024;301:118003.
[138] Hu J, Zhou H, Zhou Y, Zhang H, Nordströmd L, Yang G. Flexibility prediction of aggregated electric vehicles and domestic hot water systems in smart grids. Engineering 2021;7(8):1101–14.
[139] Zhou L, Song A. Electrification and hydrogenation on a PV-battery-hydrogen energy flexible community for carbon–neutral transformation with transient aging and collaboration operation. Energy Convers Manag 2024;300:117984.
[140] Guelpa E, Verda V. Thermal energy storage in district heating and cooling systems: a review. Appl Energy 2019. https://doi.org/10.1016/j.apenergy.2019.113474.
[141] Zhou Y, Lund PD. Peer-to-peer energy sharing and trading of renewable energy in smart communities — trading pricing models, decision-making and agent-based collaboration. Renew Energy 2023;207:177–93.
[142] Mishra AK, Jokisalo J, Kosonen R, Kinnunen T, Ekkerhaugen M, Ihasalo H, Martin K. Demand response events in district heating: results from field tests in a university building. Sustain Cities Soc 2019. https://doi.org/10.1016/j.scs.2019.101481.
[143] Guelpa E, Marincioni L, Deputato S, Capone M, Amelio S, Pochettino E, Verda V. Demand side management in district heating networks: a real application. Energy 2019;182:433–42.
[144] Guelpa E, Marincioni L. Demand side management in district heating systems by innovative control. Energy 2019. https://doi.org/10.1016/j.energy.2019.116037.
[145] Wang H, Wang S, Tang R. Development of grid-responsive buildings: opportunities, challenges, capabilities and applications of HVAC systems in non-residential buildings in providing ancillary services by fast demand responses to smart grids. Appl Energy 2019;250:697–712.
[146] Zhou Y, Cao S. Energy flexibility investigation of advanced grid-responsive energy control strategies with the static battery and electric vehicles: a case study of a high-rise office building in Hong Kong. Energy Convers Manag 2019. https://doi.org/10.1016/j.enconman.2019.111888.
[147] Lu S, Gu W, Zhou J, Zhang X, Wu C. Coordinated dispatch of multi-energy system with district heating network: modeling and solution strategy. Energy 2018;152:358–70.
[148] Pakere I, Lauka D, Blumberga D. Solar power and heat production via photovoltaic thermal panels for district heating and industrial plant. Energy 2018;154:424–32.
[149] Eicker U, Pietruschka D, Schmitt A, Haag M. Comparison of photovoltaic and solar thermal cooling systems for office buildings in different climates. Sol Energy 2015;118:243–55.
[150] Prieto A, Knaack U, Auer T, Klein T. Solar coolfacades: framework for the integration of solar cooling technologies in the building envelope. Energy 2017;137:353–68.
[151] Murugan S, Horák B. A review of micro combined heat and power systems for residential applications. Renew Sustain Energy Rev 2016;64:144–62.
[152] Rad FM, Fung AS. Solar community heating and cooling system with borehole thermal energy storage – review of systems. Renew Sustain Energy Rev 2016;60:1550–61.
[153] Gang W, Wang S, Xiao F, Gao D. District cooling systems: technology integration, system optimization, challenges and opportunities for applications. Renew Sustain Energy Rev 2016;53:253–64.
[154] Udomsri S, Martin AR, Martin V. Thermally driven cooling coupled with municipal solid waste-fired power plant: application of combined heat, cooling and power in tropical urban areas. Appl Energy 2011;88:1532–42.
[155] Meunier F. Adsorption heat powered heat pumps. Appl Therm Eng 2013;61(2):830–6.
[156] Mouchaham G, Cui FS, Nouar F, Pimenta V, Chang JS, Serre C. Metal–Organic frameworks and water: 'from old enemies to friends'? Trends in Chemistry 2020;2(11):990–1003.
[157] Boopathi Raja V, Shanmugam V. A review and new approach to minimize the cost of solar assisted absorption cooling system. Renew Sustain Energy Rev 2012;16:6725–31.
[158] Zhu LQ, Tso CY, Chan KC, Wu CL, Chao CYH, Chen J, He W, Luo SW. Experimental investigation on composite adsorbent–Water pair for a solar-powered adsorption cooling system. Appl Therm Eng 2018;131:649–59.
[159] Hassan HZ, Mohamad AA. A review on solar cold production through absorption technology. Renew Sustain Energy Rev 2012;16:5331–48.
[160] Allouhi A, Kousksou T, Jamil A, Bruel P, Mourad Y, Zeraouli Y. Solar driven cooling systems: an updated review. Renew Sustain Energy Rev 2015;44:159–81.
[161] Liu Z, Zhang L, Gong G, Han T. Experimental evaluation of an active solar thermoelectric radiant wall system. Energy Convers Manag 2015;94(0):253–60.
[162] Ibáñez-Puy M, Sacristán Fernández JA, Martín-Gómez C, Vidaurre-Arbizu M. Development and construction of a thermoelectric active facade module. J Facade Des Eng 2015;3(1):15–25.
[163] Daghigh R, Khaledian Y. Effective design, theoretical and experimental assessment of a solar thermoelectric cooling-heating system. Sol Energy 2018;162:561–72.
[164] Tugcu A, Arslan O. Optimization of geothermal energy aided absorption refrigeration system—gaars: a novel ANN-based approach. Geothermics 2017;65:210–21.

[165] Rogowska A. District cooling by a geothermal heat source. United Nations Univ; 2003. p. 465–87.
[166] Yilmaz C. Thermodynamic and economic investigation of geothermal powered absorption cooling system for buildings. Geothermics 2017;70:239–48.
[167] Chatenay C. Heat pumps and geothermal space cooling. 2014. p. 1–8.
[168] Carotenuto A, Figaj RD, Vanoli L. A novel solar-geothermal district heating, cooling and domestic hot water system: dynamic simulation and energy-economic analysis. Energy 2017;141:2652–69.
[169] Tsai JH, Wu CP, Chang HC. An investigation of geothermal energy applications and assisted air-conditioning system for energy conservation analysis. Geothermics 2014;50:220–6.
[170] Udomsri S, Bales C, Martin AR, Martin V. Decentralised cooling in district heating network: monitoring results and calibration of simulation model. Energy Build 2011;43:3311–21.
[171] Hamdy M, Askalany AA, Harby K, Kora N. An overview on adsorption cooling systems powered by waste heat from internal combustion engine. Renew Sustain Energy Rev 2015;51:1223–34.
[172] Qi X, Liu Y, Guo Q, Yu J, Yu S. Performance prediction of seawater shower cooling towers. Energy 2016;97:435–43.
[173] Hunt JD, Byers E, Sánchez AS. Technical potential and cost estimates for seawater air conditioning. Energy 2019;166:979–88.
[174] Hernández-Romero IM, Fuentes-Cortés LF, Mukherjee R, El-Halwagi MM, Serna-González M, Nápoles-Rivera F. Multi-scenario model for optimal design of seawater air-conditioning systems under demand uncertainty. J Clean Prod 2019. https://doi.org/10.1016/j.jclepro.2019.117863.
[175] Moretti E, Bonamente E, Buratti C, Cotana F. Development of innovative heating and cooling systems using renewable energy sources for non-residential buildings. Energies 2013;6:5114–29.
[176] Li Y, Rezgui Y, Zhu H. District heating and cooling optimization and enhancement – towards integration of renewables, storage and smart grid. Renew Sustain Energy Rev 2017;72:281–94.
[177] Yang L, Entchev E, Rosato A, Sibilio S. Smart thermal grid with integration of distributed and centralized solar energy systems. Energy 2017;122:471–81.
[178] Ge TS, Dai YJ, Wang RZ. Review on solar powered rotary desiccant wheel cooling system. Renew Sustain Energy Rev 2014;39:476–97.
[179] Guo J, Lin S, Bilbao JI, White SD, Sproul AB. A review of photovoltaic thermal (PV/T) heat utilisation with low temperature desiccant cooling and dehumidification. Renew Sustain Energy Rev 2017;67:1–14.
[180] Yau YH, Rismanchi B. A review on cool thermal storage technologies and operating strategies. Renew Sustain Energy Rev 2012;16(1):787–97.
[181] Yan B, Xue S, Li Y, Duan J, Zeng M. Gas-fired combined cooling, heating and power (CCHP) in Beijing: A techno-economic analysis. Renew Sustain Energy Rev 2016;63:118–31.
[182] Boait PJ, Greenough R. Can fuel cell micro-CHP justify the hydrogen gas grid? Operating experience from a UK domestic retrofit. Energy Build 2019;194:75–84.
[183] Dominković DF, Rashid KABA, Romagnoli A, Pedersen AS, Leong KC, Krajačić G, Duić N. Potential of district cooling in hot and humid climates. Appl Energy 2017;208:49–61.
[184] Absorption Chillers for CHP Systems. https://www.energy.gov/sites/prod/files/2017/06/f35/CHP-Absorption%20Chiller-compliant.pdf.
[185] Wang S, Lee JS, Wahiduzzaman M, et al. A robust large-pore zirconium carboxylate metal–organic framework for energy-efficient water-sorption-driven refrigeration. Nat Energy 2018;3:985–93.
[186] Cho KH, Borges DD, Lee UH, et al. Rational design of a robust aluminum metal-organic framework for multi-purpose water-sorption-driven heat allocations. Nat Commun 2020;11:5112.
[187] Liu J, Yang H, Zhou Y. Peer-to-peer trading optimizations on net-zero energy communities with energy storage of hydrogen and battery vehicles. Appl Energy 2021;302:117578.
[188] Zhou L. Study on thermo-electric-hydrogen conversion mechanisms and synergistic operation on hydrogen fuel cell and electrochemical battery in energy flexible buildingsEnergy Conversion and. Management 2023;277:116610.
[189] Jasiūnas J, Lund PD, Mikkola J. Energy system resilience – a review. Renew Sustain Energy Rev 2021;150:111476.
[190] Peltokorpi A, Talmar M, Castrén K, Holmström J. Designing an organizational system for economically sustainable demand-side management in district heating and cooling. J Clean Prod 2019;219:433–42.
[191] Sameti M, Haghighat F. Hybrid solar and heat-driven district cooling system: optimal integration and control strategy. Sol Energy 2019;183:260–75.
[192] Gao J, Kang J, Zhang C, Gang W. Energy performance and operation characteristics of distributed energy systems with district cooling systems in subtropical areas under different control strategies. Energy 2018;153:849–60.
[193] Sameti M, Haghighat F. Optimization approaches in district heating and cooling thermal network. Energy Build 2017;140:121–30.
[194] Cox SJ, Kim D, Cho H, Mago P. Real time optimal control of district cooling system with thermal energy storage using neural networks. Appl Energy 2019;238:466–80.
[195] Ma T, Li Z, Zhao J. Photovoltaic panel integrated with phase change materials (PV-PCM): technology overview and materials selection. Renew Sustain Energy Rev 2019. https://doi.org/10.1016/j.rser.2019.109406.
[196] Lin Y, Jia Y, Alva G, Fang G. Review on thermal conductivity enhancement, thermal properties and applications of phase change materials in thermal energy storage. Renew Sustain Energy Rev 2018;82:2730–42.
[197] DeLovato N, Sundarnath K, Cvijovic L, Kota K, Kuravi S. A review of heat recovery applications for solar and geothermal power plants. Renew Sustain Energy Rev 2019. https://doi.org/10.1016/j.rser.2019.109329.
[198] Dong LS. Ocean compressed air energy storage (OCAES) integrated with offshore renewable energy sources (PDF). Retrieved June 6, 2015..
[199] Bolund B, Bernhoff H, Leijon M. Flywheel energy and power storage systems. Renew Sustain Energy Rev 2007;2(11):235–58.
[200] United States Department of Energy. https://www.energy.gov/science-innovation.
[201] Li N, Zhang J, Zhang S, Hou X, Liu Y. The influence of accessory energy consumption on evaluation method of braking energy recovery contribution rate. Energy Convers Manag 2018;166:545–55.
[202] Askeland K, Bozhkova KN, Sorknæs P. Balancing Europe: can district heating affect the flexibility potential of Norwegian hydropower resources? Renew Energy 2019;141:646–56.
[203] Zhou Y, Cao S, Hensen JLM, Lund PD. Energy integration and interaction between buildings and vehicles: a state-of-the-art review. Renew Sustain Energy Rev 2019. https://doi.org/10.1016/j.rser.2019.109337.
[204] Kazagic A, Merzic A, Redzic E, Tresnjo D. Optimization of modular district heating solution based on CHP and RES - demonstration case of the Municipality of Visoko. Energy 2019;181:56–65.
[205] Luo X, Liu Y, Liu J, Liu X. Optimal design and cost allocation of a distributed energy resource (DER) system with district energy networks: a case study of an isolated island in the South China Sea. Sustain Cities Soc 2019. https://doi.org/10.1016/j.scs.2019.101726.
[206] Fazlollahi S, Becker G, Ashouri A, Maréchal F. Multi-objective, multi-period optimization of district energy systems: IV – a case study. Energy 2015;84:365–81.
[207] Heijde B, Vandermeulen A, Salenbien R, Helsen L. Representative days selection for district energy system optimisation: a solar district heating system with seasonal storage. Appl Energy 2019. https://doi.org/10.1016/j.apenergy.2019.04.030.
[208] Ikeda S, Ooka R. Application of differential evolution-based constrained optimization methods to district energy optimization and comparison with dynamic programming. Appl Energy 2019. https://doi.org/10.1016/j.apenergy.2019.113670.
[209] Ahmad T, Chen H. Potential of three variant machine-learning models for forecasting district level medium-term and long-term energy demand in smart grid environment. Energy 2018;160:1008–20.
[210] Zheng S, Zhou Y. Numerical study on the thermal and optical performances of an aerogel glazing system with the multivariable optimization using an advanced machine learning algorithm. Advanced Theory and Simulations 2019. https://doi.org/10.1002/adts.201900092.
[211] Manfren M, James PAB, Tronchin L. Data-driven building energy modelling – an analysis of the potential for generalisation through interpretable machine learning. Renew Sustain Energy Rev 2022;167:112686.
[212] Hwang JK, Yun GY, Lee S, Seo H, Santamouris M. Using deep learning approaches with variable selection process to predict the energy performance of a heating and cooling system. Renew Energy 2020;149:1227–45.
[213] Inderwildi O, Zhang C, Wang X, Kraft M. The impact of intelligent cyber-physical systems on the decarbonization of energy. Energy Environ Sci 2020;13:744–71.
[214] Zheng X, Zhou Y. A three-dimensional unsteady numerical model on a novel aerogel-based PV/T-PCM system with dynamic heat-transfer mechanism and solar energy harvesting analysis. Appl Energy 2023;338:120899.
[215] Zhou Y. A regression learner-based approach for battery cycling ageing prediction–advances in energy management strategy and techno-economic analysis. Energy 2022;256:124668.
[216] Zhou Y, Zheng S. Uncertainty study on thermal and energy performances of a deterministic parameters based optimal aerogel glazing system using machine-learning method. Energy 2020;193:116718.
[217] Zhou Y, Zheng S. Machine-learning based hybrid demand-side controller for high-rise office buildings with high energy flexibilities. Appl Energy 2020;262:114416.
[218] Zhou Y, Zheng S, Zhang G. Machine learning-based optimal design of a phase change material integrated renewable system with on-site PV, radiative cooling and hybrid ventilations—study of modelling and application in five climatic regions. Energy 2020;192:116608.
[219] Zhou Y, Zheng S. Machine learning-based multi-objective optimisation of an aerogel glazing system using NSGA-II—study of modelling and application in the subtropical climate Hong Kong. J Clean Prod 2020;253:119964.